

Commercial Solar Financial Analysis

The complete set — 50 kWp to 950 kWp on APSPDCL commercial tariffs.

19

STANDARD PLANT SIZES

59

PAGES OF ANALYSIS

30 yrs

POST-TAX PROJECTION EACH

50 kWp

PRICE

₹46.00 / Wp

PAYBACK

3.2 yrs

950 kWp

PRICE

₹38.00 / Wp

PAYBACK

3.5 yrs

WHAT IS IN HERE

Every capacity gets the same three pages: **returns, plant and the depreciation schedule**; then the **30-year post-tax cash flow**, year by year; then the **return metrics and the full list of assumptions** the figures rest on. Page 2 puts all 19 side by side, so the nearest size to a customer's load is one glance away.

HOW TO USE IT

Find the capacity on page 2, turn to its three pages, and read the assumptions page before you quote anything from it. These are **standard plants at list pricing**, built to show a buyer the shape of the return — not a site-specific proposal. For that, run the site through the Quote Generator.

THIS IS NOT A QUOTATION

Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the sanction APSPDCL actually grants. Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date below and are subject to change by APERC and by the Finance Act. Nothing here is tax advice — the buyer's chartered accountant confirms the tax position before anyone relies on it.

Reading the ladder. Price per watt-peak falls as capacity rises, so the larger plants pay back faster on the same tariff — and the step down from the LT tariff to the HT tariff at 150 kW cuts the value of every unit generated, which is why payback lengthens across that boundary before improving again. Levelised cost is what a unit from the plant costs over 30 years, against the grid rate in the section heading above it. The last column is where that plant's three pages begin.

CAPACITY	₹ / WP incl. GST	PRICE incl. GST	NET CAPEX after ITC	YEAR 1 units (kWh)	YEAR 1 gross saving	PAYBACK post-tax	IRR unlevered	₹ / KWH levelised	PAGE
LT CATEGORY II — COMMERCIAL · LT SUPPLY · ₹10.15 PER UNIT									
50 kWp	46.00	₹23.00 L	₹21.90 L	76,650	₹7.21 L	3.2 yrs	31.3%	3.52	3
100 kWp	46.00	₹46.00 L	₹43.81 L	1,53,300	₹14.43 L	3.2 yrs	31.3%	3.52	6
150 kWp	43.00	₹64.50 L	₹61.43 L	2,29,950	₹21.64 L	3.0 yrs	33.2%	3.32	9
HT CATEGORY I / II AT 11 KV · HT SUPPLY AT 11 KV · ₹7.65 PER UNIT									
200 kWp	43.00	₹86.00 L	₹81.90 L	3,06,600	₹21.95 L	3.9 yrs	26.2%	3.32	12
250 kWp	43.00	₹1.07 Cr	₹1.02 Cr	3,83,250	₹27.44 L	3.9 yrs	26.2%	3.32	15
300 kWp	40.50	₹1.22 Cr	₹1.16 Cr	4,59,900	₹32.93 L	3.7 yrs	27.5%	3.15	18
350 kWp	40.50	₹1.42 Cr	₹1.35 Cr	5,36,550	₹38.42 L	3.7 yrs	27.5%	3.15	21
400 kWp	40.50	₹1.62 Cr	₹1.54 Cr	6,13,200	₹43.91 L	3.7 yrs	27.5%	3.15	24
450 kWp	40.50	₹1.82 Cr	₹1.74 Cr	6,89,850	₹49.39 L	3.7 yrs	27.5%	3.15	27
500 kWp	40.50	₹2.02 Cr	₹1.93 Cr	7,66,500	₹54.88 L	3.7 yrs	27.5%	3.15	30
550 kWp	38.00	₹2.09 Cr	₹1.99 Cr	8,43,150	₹60.37 L	3.5 yrs	29.0%	2.98	33
600 kWp	38.00	₹2.28 Cr	₹2.17 Cr	9,19,800	₹65.86 L	3.5 yrs	29.0%	2.98	36
650 kWp	38.00	₹2.47 Cr	₹2.35 Cr	9,96,450	₹71.35 L	3.5 yrs	29.0%	2.98	39
700 kWp	38.00	₹2.66 Cr	₹2.53 Cr	10,73,100	₹76.83 L	3.5 yrs	29.0%	2.98	42
750 kWp	38.00	₹2.85 Cr	₹2.71 Cr	11,49,750	₹82.32 L	3.5 yrs	29.0%	2.98	45
800 kWp	38.00	₹3.04 Cr	₹2.90 Cr	12,26,400	₹87.81 L	3.5 yrs	29.0%	2.98	48
850 kWp	38.00	₹3.23 Cr	₹3.08 Cr	13,03,050	₹93.30 L	3.5 yrs	29.0%	2.98	51
900 kWp	38.00	₹3.42 Cr	₹3.26 Cr	13,79,700	₹98.79 L	3.5 yrs	29.0%	2.98	54
950 kWp	38.00	₹3.61 Cr	₹3.44 Cr	14,56,350	₹1.04 Cr	3.5 yrs	29.0%	2.98	57

What these figures exclude. Demand and fixed charges, which solar does not reduce. Financing, so every return is unlevered — the return on the money, before any loan. Inverter replacement, which a real thirty-year plant will need once. Each deck's third page lists the assumptions in full, and they are the same for all 19.

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

50 kWp

PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

50 KWP GRID-TIED SOLAR PLANT

LT CATEGORY II — COMMERCIAL

Net investment ₹21.90 L after input tax credit, generating 76,650 units in year one against an energy charge of ₹10.15 per unit.

30-YEAR ANALYSIS

PAYBACK

3.2 yrs

Post-tax, undiscounted

PROJECT IRR

31.3%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.52

per kWh, against ₹10.15 from the grid

NET PRESENT VALUE

₹50.14 L

Discounted at 10%

PLANT AND GENERATION

Plant capacity	50 kWp
Year 1 generation	76,650 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	LT Category II — Commercial (LT supply)
Energy charge offset	₹10.15 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹7.21 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹46.00 / Wp (incl. GST)
Project price incl. GST	₹23.00 L
GST at 5% — recovered as ITC	₹1.10 L
Net capitalised cost	₹21.90 L
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹5.51 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	21,90,476	8,76,190	13,14,286	2,20,519
2	13,14,285	5,25,714	7,88,571	1,32,312
3	7,88,572	3,15,429	4,73,143	79,387
4	4,73,143	1,89,257	2,83,886	47,632
5	2,83,885	1,13,554	1,70,331	28,579
6	1,70,332	68,133	1,02,199	17,148
7	1,02,199	40,880	61,319	10,289
8	61,320	24,528	36,792	6,173
9	36,792	14,717	22,075	3,704
10	22,075	8,830	13,245	2,222
11	13,245	5,298	7,947	1,333
12	7,947	3,179	4,768	800
Tax shield, years 1-12				5,50,099

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

50 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	76,650	7,21,277	—	8,76,190	(38,989)	7,60,265	7,60,265
2	76,267	7,46,377	30,000	5,25,714	47,986	6,68,391	14,28,656
3	75,885	7,72,351	30,000	3,15,429	1,07,448	6,34,903	20,63,559
4	75,506	7,99,229	30,000	1,89,257	1,45,967	6,23,261	26,86,820
5	75,128	8,27,042	30,000	1,13,554	1,72,020	6,25,022	33,11,842
6	74,753	8,55,823	30,000	68,133	1,90,695	6,35,127	39,46,969
7	74,379	8,85,605	30,000	40,880	2,05,050	6,50,555	45,97,524
8	74,007	9,16,425	30,000	24,528	2,16,922	6,69,502	52,67,026
9	73,637	9,48,316	30,000	14,717	2,27,418	6,90,898	59,57,924
10	73,269	9,81,318	30,000	8,830	2,37,205	7,14,112	66,72,036
11	72,903	10,15,467	30,000	5,298	2,46,689	7,38,778	74,10,814
12	72,538	10,50,806	30,000	3,179	2,56,116	7,64,689	81,75,503
13	72,175	10,87,374	30,000	1,907	2,65,640	7,91,734	89,67,237
14	71,815	11,25,214	30,000	1,144	2,75,356	8,19,859	97,87,096
15	71,455	11,64,372	30,000	687	2,85,326	8,49,046	1,06,36,142
16	71,098	12,04,892	30,000	412	2,95,593	8,79,299	1,15,15,441
17	70,743	12,46,822	30,000	247	3,06,188	9,10,635	1,24,26,076
18	70,389	12,90,212	30,000	148	3,17,133	9,43,079	1,33,69,155
19	70,037	13,35,111	30,000	89	3,28,448	9,76,663	1,43,45,818
20	69,687	13,81,573	30,000	53	3,40,150	10,11,422	1,53,57,240
21	69,338	14,29,651	30,000	32	3,52,256	10,47,395	1,64,04,635
22	68,992	14,79,403	30,000	19	3,64,781	10,84,622	1,74,89,257
23	68,647	15,30,887	30,000	12	3,77,740	11,23,146	1,86,12,403
24	68,304	15,84,161	30,000	7	3,91,150	11,63,012	1,97,75,415
25	67,962	16,39,290	30,000	4	4,05,025	12,04,265	2,09,79,680
26	67,622	16,96,338	30,000	2	4,19,383	12,46,954	2,22,26,634
27	67,284	17,55,370	30,000	1	4,34,241	12,91,129	2,35,17,763
28	66,948	18,16,457	30,000	1	4,49,615	13,36,842	2,48,54,605
29	66,613	18,79,670	30,000	1	4,65,525	13,84,145	2,62,38,750
30	66,280	19,45,082	30,000	0	4,81,988	14,33,094	2,76,71,844
Total	21,40,311	3,71,11,915	8,70,000	21,90,475	85,70,065	2,76,71,844	2,76,71,844

RETURN METRICS

Simple payback	3.2 yrs
Discounted payback	4.1 yrs
Project IRR, unlevered	31.3%
Net present value at 10%	₹50.14 L

OVER THE FULL TERM

Levelised cost of energy	₹3.52 / kWh
Cumulative return, 30 years	1163%
Total net cash flow, after tax	₹2.77 Cr
30-year generation	21,40,310 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.52
Grid energy charge today		₹10.15

Every unit the plant generates costs ₹3.52 across its life against ₹10.15 bought from APSPDCL today — and the grid figure reaches ₹31.65 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 21,40,310 units, avoiding 1,755.1 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** LT Category II — Commercial at ₹10.15/kWh. Marginal rate, above 500 units per month. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹23.00 L.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 50 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

100 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

100 KWP GRID-TIED SOLAR PLANT

LT CATEGORY II — COMMERCIAL

Net investment ₹43.81 L after input tax credit, generating 1,53,300 units in year one against an energy charge of ₹10.15 per unit.

30-YEAR ANALYSIS

PAYBACK

3.2 yrs

Post-tax, undiscounted

PROJECT IRR

31.3%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.52

per kWh, against ₹10.15 from the grid

NET PRESENT VALUE

₹1.00 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	100 kWp
Year 1 generation	1,53,300 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	LT Category II — Commercial (LT supply)
Energy charge offset	₹10.15 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹14.43 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹46.00 / Wp (incl. GST)
Project price incl. GST	₹46.00 L
GST at 5% — recovered as ITC	₹2.19 L
Net capitalised cost	₹43.81 L
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹11.03 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	43,80,952	17,52,381	26,28,571	4,41,039
2	26,28,572	10,51,429	15,77,143	2,64,624
3	15,77,143	6,30,857	9,46,286	1,58,774
4	9,46,285	3,78,514	5,67,771	95,264
5	5,67,772	2,27,109	3,40,663	57,159
6	3,40,663	1,36,265	2,04,398	34,295
7	2,04,398	81,759	1,22,639	20,577
8	1,22,638	49,055	73,583	12,346
9	73,583	29,433	44,150	7,408
10	44,150	17,660	26,490	4,445
11	26,490	10,596	15,894	2,667
12	15,894	6,358	9,536	1,600
Tax shield, years 1-12				11,00,198

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

100 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	1,53,300	14,42,553	—	17,52,381	(77,977)	15,20,530	15,20,530
2	1,52,534	14,92,754	60,000	10,51,429	95,972	13,36,782	28,57,312
3	1,51,771	15,44,702	60,000	6,30,857	2,14,896	12,69,806	41,27,118
4	1,51,012	15,98,457	60,000	3,78,514	2,91,934	12,46,523	53,73,641
5	1,50,257	16,54,084	60,000	2,27,109	3,44,040	12,50,043	66,23,684
6	1,49,506	17,11,646	60,000	1,36,265	3,81,391	12,70,255	78,93,939
7	1,48,758	17,71,211	60,000	81,759	4,10,100	13,01,111	91,95,050
8	1,48,014	18,32,849	60,000	49,055	4,33,844	13,39,005	1,05,34,055
9	1,47,274	18,96,632	60,000	29,433	4,54,836	13,81,796	1,19,15,851
10	1,46,538	19,62,635	60,000	17,660	4,74,411	14,28,225	1,33,44,076
11	1,45,805	20,30,935	60,000	10,596	4,93,378	14,77,557	1,48,21,633
12	1,45,076	21,01,611	60,000	6,358	5,12,233	15,29,379	1,63,51,012
13	1,44,351	21,74,747	60,000	3,815	5,31,280	15,83,468	1,79,34,480
14	1,43,629	22,50,429	60,000	2,289	5,50,711	16,39,718	1,95,74,198
15	1,42,911	23,28,744	60,000	1,373	5,70,652	16,98,092	2,12,72,290
16	1,42,196	24,09,784	60,000	824	5,91,186	17,58,598	2,30,30,888
17	1,41,485	24,93,644	60,000	494	6,12,375	18,21,269	2,48,52,157
18	1,40,778	25,80,423	60,000	297	6,34,265	18,86,158	2,67,38,315
19	1,40,074	26,70,222	60,000	178	6,56,896	19,53,326	2,86,91,641
20	1,39,374	27,63,146	60,000	107	6,80,301	20,22,845	3,07,14,486
21	1,38,677	28,59,303	60,000	64	7,04,512	20,94,791	3,28,09,277
22	1,37,983	29,58,807	60,000	38	7,29,562	21,69,245	3,49,78,522
23	1,37,293	30,61,773	60,000	23	7,55,480	22,46,293	3,72,24,815
24	1,36,607	31,68,323	60,000	14	7,82,299	23,26,024	3,95,50,839
25	1,35,924	32,78,581	60,000	8	8,10,050	24,08,530	4,19,59,369
26	1,35,244	33,92,675	60,000	5	8,38,766	24,93,909	4,44,53,278
27	1,34,568	35,10,740	60,000	3	8,68,482	25,82,259	4,70,35,537
28	1,33,895	36,32,914	60,000	2	8,99,231	26,73,683	4,97,09,220
29	1,33,226	37,59,339	60,000	1	9,31,049	27,68,290	5,24,77,510
30	1,32,560	38,90,164	60,000	1	9,63,976	28,66,189	5,53,43,699
Total	42,80,620	7,42,23,827	17,40,000	43,80,952	1,71,40,131	5,53,43,699	5,53,43,699

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

100 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.2 yrs
Discounted payback	4.1 yrs
Project IRR, unlevered	31.3%
Net present value at 10%	₹1.00 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.52 / kWh
Cumulative return, 30 years	1163%
Total net cash flow, after tax	₹5.53 Cr
30-year generation	42,80,621 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.52
Grid energy charge today		₹10.15

Every unit the plant generates costs ₹3.52 across its life against ₹10.15 bought from APSPDCL today — and the grid figure reaches ₹31.65 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 42,80,621 units, avoiding 3,510.1 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** LT Category II — Commercial at ₹10.15/kWh. Marginal rate, above 500 units per month. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹46.00 L.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 100 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

150 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

150 KWP GRID-TIED SOLAR PLANT

LT CATEGORY II — COMMERCIAL

Net investment ₹61.43 L after input tax credit, generating 2,29,950 units in year one against an energy charge of ₹10.15 per unit.

30-YEAR ANALYSIS

PAYBACK

3.0 yrs

Post-tax, undiscounted

PROJECT IRR

33.2%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.32

per kWh, against ₹10.15 from the grid

NET PRESENT VALUE

₹1.54 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	150 kWp
Year 1 generation	2,29,950 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	LT Category II — Commercial (LT supply)
Energy charge offset	₹10.15 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹21.64 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹43.00 / Wp (incl. GST)
Project price incl. GST	₹64.50 L
GST at 5% — recovered as ITC	₹3.07 L
Net capitalised cost	₹61.43 L
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹15.46 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	61,42,857	24,57,143	36,85,714	6,18,414
2	36,85,715	14,74,286	22,11,429	3,71,048
3	22,11,428	8,84,571	13,26,857	2,22,629
4	13,26,857	5,30,743	7,96,114	1,33,577
5	7,96,115	3,18,446	4,77,669	80,146
6	4,77,668	1,91,067	2,86,601	48,088
7	2,86,601	1,14,640	1,71,961	28,853
8	1,71,960	68,784	1,03,176	17,312
9	1,03,177	41,271	61,906	10,387
10	61,906	24,762	37,144	6,232
11	37,143	14,857	22,286	3,739
12	22,286	8,914	13,372	2,243
Tax shield, years 1-12				15,42,669

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

150 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	2,29,950	21,63,830	—	24,57,143	(73,821)	22,37,651	22,37,651
2	2,28,800	22,39,131	90,000	14,74,286	1,69,845	19,79,286	42,16,937
3	2,27,656	23,17,053	90,000	8,84,571	3,37,876	18,89,177	61,06,114
4	2,26,518	23,97,686	90,000	5,30,743	4,47,221	18,60,465	79,66,579
5	2,25,385	24,81,125	90,000	3,18,446	5,21,652	18,69,473	98,36,052
6	2,24,258	25,67,469	90,000	1,91,067	5,75,441	19,02,027	1,17,38,079
7	2,23,137	26,56,816	90,000	1,14,640	6,17,164	19,49,653	1,36,87,732
8	2,22,021	27,49,274	90,000	68,784	6,51,974	20,07,299	1,56,95,031
9	2,20,911	28,44,948	90,000	41,271	6,82,978	20,71,970	1,77,67,001
10	2,19,807	29,43,953	90,000	24,762	7,12,051	21,41,902	1,99,08,903
11	2,18,708	30,46,402	90,000	14,857	7,40,328	22,16,074	2,21,24,977
12	2,17,614	31,52,417	90,000	8,914	7,68,506	22,93,911	2,44,18,888
13	2,16,526	32,62,121	90,000	5,349	7,97,013	23,75,108	2,67,93,996
14	2,15,444	33,75,643	90,000	3,209	8,26,123	24,59,520	2,92,53,516
15	2,14,366	34,93,115	90,000	1,926	8,56,011	25,47,104	3,18,00,620
16	2,13,294	36,14,676	90,000	1,155	8,86,800	26,37,876	3,44,38,496
17	2,12,228	37,40,466	90,000	693	9,18,575	27,31,891	3,71,70,387
18	2,11,167	38,70,635	90,000	416	9,51,405	28,29,229	3,99,99,616
19	2,10,111	40,05,333	90,000	250	9,85,348	29,29,985	4,29,29,601
20	2,09,060	41,44,718	90,000	150	10,20,454	30,34,264	4,59,63,865
21	2,08,015	42,88,954	90,000	90	10,56,770	31,42,184	4,91,06,049
22	2,06,975	44,38,210	90,000	54	10,94,344	32,53,866	5,23,59,915
23	2,05,940	45,92,660	90,000	32	11,33,221	33,69,439	5,57,29,354
24	2,04,911	47,52,484	90,000	19	11,73,449	34,89,035	5,92,18,389
25	2,03,886	49,17,871	90,000	12	12,15,076	36,12,795	6,28,31,184
26	2,02,867	50,89,013	90,000	7	12,58,150	37,40,863	6,65,72,047
27	2,01,852	52,66,110	90,000	4	13,02,722	38,73,388	7,04,45,435
28	2,00,843	54,49,371	90,000	3	13,48,846	40,10,525	7,44,55,960
29	1,99,839	56,39,009	90,000	2	13,96,574	41,52,435	7,86,08,395
30	1,98,840	58,35,247	90,000	1	14,45,963	42,99,283	8,29,07,678
Total	64,20,929	11,13,35,740	26,10,000	61,42,856	2,58,18,059	8,29,07,678	8,29,07,678

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

150 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.0 yrs
Discounted payback	3.8 yrs
Project IRR, unlevered	33.2%
Net present value at 10%	₹1.54 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.32 / kWh
Cumulative return, 30 years	1250%
Total net cash flow, after tax	₹8.29 Cr
30-year generation	64,20,931 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.32
Grid energy charge today		₹10.15

Every unit the plant generates costs **₹3.32** across its life against **₹10.15** bought from APSPDCL today — and the grid figure reaches **₹31.65** by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 64,20,931 units, avoiding 5,265.2 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** LT Category II — Commercial at ₹10.15/kWh. Marginal rate, above 500 units per month. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹64.50 L.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 150 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

APSPDCL · COMMERCIAL ROOFTOP

200 KWP GRID-TIED SOLAR PLANT

Net investment ₹81.90 L after input tax credit, generating 3,06,600 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK 3.9 yrs Post-tax, undiscounted	PROJECT IRR 26.2% Unlevered, 30 years	COST OF SOLAR ENERGY ₹3.32 per kWh, against ₹7.65 from the grid	NET PRESENT VALUE ₹1.39 Cr Discounted at 10%
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PLANT AND GENERATION		INVESTMENT AND TAX BASIS	
Plant capacity	200 kWp	Price per watt-peak	₹43.00 / Wp (incl. GST)
Year 1 generation	3,06,600 kWh	Project price incl. GST	₹86.00 L
Specific yield	1,533 kWh / kWp / year	GST at 5% — recovered as ITC	₹4.10 L
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)	Net capitalised cost	₹81.90 L
Energy charge offset	₹7.65 / kWh	Depreciation block	40% WDV, Appendix I
Units self-consumed	90% at the energy charge	Tax regime	s.115BAA — 25.168%
Units exported	10% at ₹2.75 / kWh (APPC)	Annual maintenance	₹600 / kWp / year, year 1 free
Year 1 gross saving	₹21.95 L	30-year depreciation shield	₹20.61 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	81,90,476	32,76,190	49,14,286	8,24,551
2	49,14,285	19,65,714	29,48,571	4,94,731
3	29,48,572	11,79,429	17,69,143	2,96,839
4	17,69,143	7,07,657	10,61,486	1,78,103
5	10,61,485	4,24,594	6,36,891	1,06,862
6	6,36,892	2,54,757	3,82,135	64,117
7	3,82,135	1,52,854	2,29,281	38,470
8	2,29,281	91,712	1,37,569	23,082
9	1,37,568	55,027	82,541	13,849
10	82,541	33,016	49,525	8,309
11	49,525	19,810	29,715	4,986
12	29,715	11,886	17,829	2,991
Tax shield, years 1-12				20,56,892

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

200 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	3,06,600	21,95,256	—	32,76,190	(2,72,050)	24,67,306	24,67,306
2	3,05,067	22,71,651	1,20,000	19,65,714	46,797	21,04,854	45,72,160
3	3,03,542	23,50,704	1,20,000	11,79,429	2,64,585	19,66,119	65,38,279
4	3,02,024	24,32,509	1,20,000	7,07,657	4,03,909	19,08,600	84,46,879
5	3,00,514	25,17,160	1,20,000	4,24,594	4,96,455	19,00,705	1,03,47,584
6	2,99,011	26,04,757	1,20,000	2,54,757	5,61,247	19,23,511	1,22,71,095
7	2,97,516	26,95,403	1,20,000	1,52,854	6,09,707	19,65,696	1,42,36,791
8	2,96,029	27,89,203	1,20,000	91,712	6,48,703	20,20,500	1,62,57,291
9	2,94,548	28,86,267	1,20,000	55,027	6,82,365	20,83,902	1,83,41,193
10	2,93,076	29,86,709	1,20,000	33,016	7,13,184	21,53,525	2,04,94,718
11	2,91,610	30,90,647	1,20,000	19,810	7,42,667	22,27,980	2,27,22,698
12	2,90,152	31,98,201	1,20,000	11,886	7,71,730	23,06,471	2,50,29,169
13	2,88,702	33,09,499	1,20,000	7,132	8,00,938	23,88,561	2,74,17,730
14	2,87,258	34,24,669	1,20,000	4,279	8,30,642	24,74,027	2,98,91,757
15	2,85,822	35,43,848	1,20,000	2,567	8,61,068	25,62,780	3,24,54,537
16	2,84,393	36,67,174	1,20,000	1,540	8,92,365	26,54,809	3,51,09,346
17	2,82,971	37,94,791	1,20,000	924	9,24,639	27,50,152	3,78,59,498
18	2,81,556	39,26,850	1,20,000	555	9,57,968	28,48,882	4,07,08,380
19	2,80,148	40,63,504	1,20,000	333	9,92,417	29,51,087	4,36,59,467
20	2,78,747	42,04,914	1,20,000	200	10,28,041	30,56,873	4,67,16,340
21	2,77,354	43,51,245	1,20,000	120	10,64,890	31,66,356	4,98,82,696
22	2,75,967	45,02,669	1,20,000	72	11,03,012	32,79,657	5,31,62,353
23	2,74,587	46,59,362	1,20,000	43	11,42,456	33,96,906	5,65,59,259
24	2,73,214	48,21,507	1,20,000	26	11,83,269	35,18,238	6,00,77,497
25	2,71,848	49,89,296	1,20,000	16	12,25,500	36,43,795	6,37,21,292
26	2,70,489	51,62,923	1,20,000	9	12,69,201	37,73,723	6,74,95,015
27	2,69,136	53,42,593	1,20,000	6	13,14,421	39,08,172	7,14,03,187
28	2,67,791	55,28,515	1,20,000	3	13,61,214	40,47,301	7,54,50,488
29	2,66,452	57,20,908	1,20,000	2	14,09,636	41,91,272	7,96,41,760
30	2,65,119	59,19,995	1,20,000	1	14,59,742	43,40,253	8,39,82,013
Total	85,61,243	11,29,52,729	34,80,000	81,90,474	2,54,90,718	8,39,82,013	8,39,82,013

RETURN METRICS

Simple payback	3.9 yrs
Discounted payback	5.2 yrs
Project IRR, unlevered	26.2%
Net present value at 10%	₹1.39 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.32 / kWh
Cumulative return, 30 years	925%
Total net cash flow, after tax	₹8.40 Cr
30-year generation	85,61,241 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.32
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.32 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 85,61,241 units, avoiding 7,020.2 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹86.00 L.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 200 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

APSPDCL · COMMERCIAL ROOFTOP

250 KWP GRID-TIED SOLAR PLANT

Net investment ₹1.02 Cr after input tax credit, generating 3,83,250 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.9 yrs

Post-tax, undiscounted

PROJECT IRR

26.2%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.32

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹1.73 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	250 kWp
Year 1 generation	3,83,250 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹27.44 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹43.00 / Wp (incl. GST)
Project price incl. GST	₹1.07 Cr
GST at 5% — recovered as ITC	₹5.12 L
Net capitalised cost	₹1.02 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹25.77 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,02,38,095	40,95,238	61,42,857	10,30,689
2	61,42,857	24,57,143	36,85,714	6,18,414
3	36,85,715	14,74,286	22,11,429	3,71,048
4	22,11,428	8,84,571	13,26,857	2,22,629
5	13,26,857	5,30,743	7,96,114	1,33,577
6	7,96,115	3,18,446	4,77,669	80,146
7	4,77,668	1,91,067	2,86,601	48,088
8	2,86,601	1,14,640	1,71,961	28,853
9	1,71,960	68,784	1,03,176	17,312
10	1,03,177	41,271	61,906	10,387
11	61,906	24,762	37,144	6,232
12	37,143	14,857	22,286	3,739
Tax shield, years 1-12				25,71,115

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	3,83,250	27,44,070	—	40,95,238	(3,40,062)	30,84,132	30,84,132
2	3,81,334	28,39,564	1,50,000	24,57,143	58,496	26,31,068	57,15,200
3	3,79,427	29,38,380	1,50,000	14,74,286	3,30,731	24,57,649	81,72,849
4	3,77,530	30,40,636	1,50,000	8,84,571	5,04,886	23,85,750	1,05,58,599
5	3,75,642	31,46,450	1,50,000	5,30,743	6,20,569	23,75,881	1,29,34,480
6	3,73,764	32,55,947	1,50,000	3,18,446	7,01,558	24,04,388	1,53,38,868
7	3,71,895	33,69,254	1,50,000	1,91,067	7,62,134	24,57,120	1,77,95,988
8	3,70,036	34,86,504	1,50,000	1,14,640	8,10,879	25,25,625	2,03,21,613
9	3,68,186	36,07,834	1,50,000	68,784	8,52,956	26,04,878	2,29,26,491
10	3,66,345	37,33,387	1,50,000	41,271	8,91,480	26,91,907	2,56,18,398
11	3,64,513	38,63,308	1,50,000	24,762	9,28,333	27,84,975	2,84,03,373
12	3,62,690	39,97,752	1,50,000	14,857	9,64,663	28,83,089	3,12,86,462
13	3,60,877	41,36,873	1,50,000	8,914	10,01,173	29,85,701	3,42,72,163
14	3,59,073	42,80,837	1,50,000	5,349	10,38,303	30,92,534	3,73,64,697
15	3,57,277	44,29,810	1,50,000	3,209	10,76,335	32,03,475	4,05,68,172
16	3,55,491	45,83,967	1,50,000	1,926	11,15,456	33,18,511	4,38,86,683
17	3,53,713	47,43,489	1,50,000	1,155	11,55,799	34,37,691	4,73,24,374
18	3,51,945	49,08,563	1,50,000	693	11,97,461	35,61,102	5,08,85,476
19	3,50,185	50,79,380	1,50,000	416	12,40,522	36,88,859	5,45,74,335
20	3,48,434	52,56,143	1,50,000	250	12,85,051	38,21,092	5,83,95,427
21	3,46,692	54,39,057	1,50,000	150	13,31,112	39,57,945	6,23,53,372
22	3,44,959	56,28,336	1,50,000	90	13,78,765	40,99,571	6,64,52,943
23	3,43,234	58,24,202	1,50,000	54	14,28,070	42,46,132	7,06,99,075
24	3,41,518	60,26,884	1,50,000	32	14,79,086	43,97,798	7,50,96,873
25	3,39,810	62,36,620	1,50,000	19	15,31,876	45,54,744	7,96,51,617
26	3,38,111	64,53,654	1,50,000	12	15,86,501	47,17,153	8,43,68,770
27	3,36,420	66,78,241	1,50,000	7	16,43,026	48,85,215	8,92,53,985
28	3,34,738	69,10,644	1,50,000	4	17,01,518	50,59,126	9,43,13,111
29	3,33,065	71,51,135	1,50,000	3	17,62,045	52,39,090	9,95,52,201
30	3,31,399	73,99,994	1,50,000	2	18,24,678	54,25,316	10,49,77,517
Total	1,07,01,553	14,11,90,915	43,50,000	1,02,38,093	3,18,63,400	10,49,77,517	10,49,77,517

RETURN METRICS

Simple payback	3.9 yrs
Discounted payback	5.2 yrs
Project IRR, unlevered	26.2%
Net present value at 10%	₹1.73 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.32 / kWh
Cumulative return, 30 years	925%
Total net cash flow, after tax	₹10.50 Cr
30-year generation	1,07,01,552 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.32
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.32 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 1,07,01,552 units, avoiding 8,775.3 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹1.07 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(iia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 250 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

APSPDCL · COMMERCIAL ROOFTOP

300 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

Net investment ₹1.16 Cr after input tax credit, generating 4,59,900 units in year one against an energy charge of ₹7.65 per unit.

30-YEAR ANALYSIS

PAYBACK

3.7 yrs

Post-tax, undiscounted

PROJECT IRR

27.5%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.15

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹2.14 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	300 kWp
Year 1 generation	4,59,900 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹32.93 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹40.50 / Wp (incl. GST)
Project price incl. GST	₹1.22 Cr
GST at 5% — recovered as ITC	₹5.79 L
Net capitalised cost	₹1.16 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹29.12 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,15,71,428	46,28,571	69,42,857	11,64,919
2	69,42,857	27,77,143	41,65,714	6,98,951
3	41,65,715	16,66,286	24,99,429	4,19,371
4	24,99,428	9,99,771	14,99,657	2,51,622
5	14,99,657	5,99,863	8,99,794	1,50,974
6	8,99,795	3,59,918	5,39,877	90,584
7	5,39,877	2,15,951	3,23,926	54,351
8	3,23,926	1,29,570	1,94,356	32,610
9	1,94,355	77,742	1,16,613	19,566
10	1,16,613	46,645	69,968	11,740
11	69,968	27,987	41,981	7,044
12	41,980	16,792	25,188	4,226
Tax shield, years 1-12				29,05,957

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

300 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	4,59,900	32,92,884	—	46,28,571	(3,36,166)	36,29,050	36,29,050
2	4,57,601	34,07,476	1,80,000	27,77,143	1,13,340	31,14,136	67,43,186
3	4,55,312	35,26,057	1,80,000	16,66,286	4,22,765	29,23,292	96,66,478
4	4,53,036	36,48,763	1,80,000	9,99,771	6,21,396	28,47,367	1,25,13,845
5	4,50,771	37,75,740	1,80,000	5,99,863	7,54,002	28,41,738	1,53,55,583
6	4,48,517	39,07,136	1,80,000	3,59,918	8,47,462	28,79,675	1,82,35,258
7	4,46,274	40,43,104	1,80,000	2,15,951	9,17,916	29,45,189	2,11,80,447
8	4,44,043	41,83,804	1,80,000	1,29,570	9,75,067	30,28,737	2,42,09,184
9	4,41,823	43,29,401	1,80,000	77,742	10,24,755	31,24,646	2,73,33,830
10	4,39,614	44,80,064	1,80,000	46,645	10,70,500	32,29,564	3,05,63,394
11	4,37,416	46,35,970	1,80,000	27,987	11,14,435	33,41,535	3,39,04,929
12	4,35,228	47,97,302	1,80,000	16,792	11,57,856	34,59,446	3,73,64,375
13	4,33,052	49,64,248	1,80,000	10,075	12,01,564	35,82,684	4,09,47,059
14	4,30,887	51,37,004	1,80,000	6,045	12,46,057	37,10,947	4,46,58,006
15	4,28,733	53,15,772	1,80,000	3,627	12,91,658	38,44,113	4,85,02,119
16	4,26,589	55,00,760	1,80,000	2,176	13,38,581	39,82,179	5,24,84,298
17	4,24,456	56,92,187	1,80,000	1,306	13,86,979	41,25,208	5,66,09,506
18	4,22,334	58,90,275	1,80,000	783	14,36,965	42,73,310	6,08,82,816
19	4,20,222	60,95,257	1,80,000	470	14,88,633	44,26,623	6,53,09,439
20	4,18,121	63,07,372	1,80,000	282	15,42,066	45,85,306	6,98,94,745
21	4,16,030	65,26,868	1,80,000	169	15,97,337	47,49,531	7,46,44,276
22	4,13,950	67,54,003	1,80,000	102	16,54,520	49,19,484	7,95,63,760
23	4,11,880	69,89,042	1,80,000	61	17,13,684	50,95,358	8,46,59,118
24	4,09,821	72,32,261	1,80,000	37	17,74,904	52,77,357	8,99,36,475
25	4,07,772	74,83,944	1,80,000	22	18,38,251	54,65,693	9,54,02,168
26	4,05,733	77,44,385	1,80,000	13	19,03,801	56,60,584	10,10,62,752
27	4,03,704	80,13,890	1,80,000	8	19,71,631	58,62,258	10,69,25,010
28	4,01,686	82,92,773	1,80,000	5	20,41,821	60,70,951	11,29,95,961
29	3,99,677	85,81,361	1,80,000	3	21,14,454	62,86,907	11,92,82,868
30	3,97,679	88,79,993	1,80,000	2	21,89,614	65,10,379	12,57,93,247
Total	1,28,41,861	16,94,29,096	52,20,000	1,15,71,425	3,84,15,848	12,57,93,247	12,57,93,247

RETURN METRICS

Simple payback	3.7 yrs
Discounted payback	4.9 yrs
Project IRR, unlevered	27.5%
Net present value at 10%	₹2.14 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.15 / kWh
Cumulative return, 30 years	987%
Total net cash flow, after tax	₹12.58 Cr
30-year generation	1,28,41,862 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.15
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.15 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 1,28,41,862 units, avoiding 10,530.3 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹1.22 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 300 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

APSPDCL · COMMERCIAL ROOFTOP

350 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

Net investment ₹1.35 Cr after input tax credit, generating 5,36,550 units in year one against an energy charge of ₹7.65 per unit.

30-YEAR ANALYSIS

PAYBACK 3.7 yrs Post-tax, undiscounted	PROJECT IRR 27.5% Unlevered, 30 years	COST OF SOLAR ENERGY ₹3.15 per kWh, against ₹7.65 from the grid	NET PRESENT VALUE ₹2.49 Cr Discounted at 10%
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PLANT AND GENERATION	INVESTMENT AND TAX BASIS																																
<table><tr><td>Plant capacity</td><td>350 kWp</td></tr><tr><td>Year 1 generation</td><td>5,36,550 kWh</td></tr><tr><td>Specific yield</td><td>1,533 kWh / kWp / year</td></tr><tr><td>Supply category</td><td>HT Category I / II at 11 kV (HT supply at 11 kV)</td></tr><tr><td>Energy charge offset</td><td>₹7.65 / kWh</td></tr><tr><td>Units self-consumed</td><td>90% at the energy charge</td></tr><tr><td>Units exported</td><td>10% at ₹2.75 / kWh (APPC)</td></tr><tr><td>Year 1 gross saving</td><td>₹38.42 L</td></tr></table>	Plant capacity	350 kWp	Year 1 generation	5,36,550 kWh	Specific yield	1,533 kWh / kWp / year	Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)	Energy charge offset	₹7.65 / kWh	Units self-consumed	90% at the energy charge	Units exported	10% at ₹2.75 / kWh (APPC)	Year 1 gross saving	₹38.42 L	<table><tr><td>Price per watt-peak</td><td>₹40.50 / Wp (incl. GST)</td></tr><tr><td>Project price incl. GST</td><td>₹1.42 Cr</td></tr><tr><td>GST at 5% — recovered as ITC</td><td>₹6.75 L</td></tr><tr><td>Net capitalised cost</td><td>₹1.35 Cr</td></tr><tr><td>Depreciation block</td><td>40% WDV, Appendix I</td></tr><tr><td>Tax regime</td><td>s.115BAA — 25.168%</td></tr><tr><td>Annual maintenance</td><td>₹600 / kWp / year, year 1 free</td></tr><tr><td>30-year depreciation shield</td><td>₹33.98 L</td></tr></table>	Price per watt-peak	₹40.50 / Wp (incl. GST)	Project price incl. GST	₹1.42 Cr	GST at 5% — recovered as ITC	₹6.75 L	Net capitalised cost	₹1.35 Cr	Depreciation block	40% WDV, Appendix I	Tax regime	s.115BAA — 25.168%	Annual maintenance	₹600 / kWp / year, year 1 free	30-year depreciation shield	₹33.98 L
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30-year depreciation shield	₹33.98 L																																

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,35,00,000	54,00,000	81,00,000	13,59,072
2	81,00,000	32,40,000	48,60,000	8,15,443
3	48,60,000	19,44,000	29,16,000	4,89,266
4	29,16,000	11,66,400	17,49,600	2,93,560
5	17,49,600	6,99,840	10,49,760	1,76,136
6	10,49,760	4,19,904	6,29,856	1,05,681
7	6,29,856	2,51,942	3,77,914	63,409
8	3,77,913	1,51,165	2,26,748	38,045
9	2,26,748	90,699	1,36,049	22,827
10	1,36,049	54,420	81,629	13,696
11	81,630	32,652	48,978	8,218
12	48,978	19,591	29,387	4,931
Tax shield, years 1-12				33,90,284

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

350 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	5,36,550	38,41,698	—	54,00,000	(3,92,193)	42,33,891	42,33,891
2	5,33,867	39,75,389	2,10,000	32,40,000	1,32,230	36,33,159	78,67,050
3	5,31,198	41,13,733	2,10,000	19,44,000	4,93,226	34,10,507	1,12,77,557
4	5,28,542	42,56,891	2,10,000	11,66,400	7,24,962	33,21,929	1,45,99,486
5	5,25,899	44,05,030	2,10,000	6,99,840	8,79,669	33,15,361	1,79,14,847
6	5,23,270	45,58,325	2,10,000	4,19,904	9,88,705	33,59,620	2,12,74,467
7	5,20,653	47,16,955	2,10,000	2,51,942	10,70,902	34,36,053	2,47,10,520
8	5,18,050	48,81,105	2,10,000	1,51,165	11,37,578	35,33,527	2,82,44,047
9	5,15,460	50,50,968	2,10,000	90,699	11,95,548	36,45,420	3,18,89,467
10	5,12,883	52,26,741	2,10,000	54,420	12,48,917	37,67,824	3,56,57,291
11	5,10,318	54,08,632	2,10,000	32,652	13,00,174	38,98,458	3,95,55,749
12	5,07,767	55,96,852	2,10,000	19,591	13,50,832	40,36,020	4,35,91,769
13	5,05,228	57,91,623	2,10,000	11,755	14,01,824	41,79,798	4,77,71,567
14	5,02,702	59,93,171	2,10,000	7,053	14,53,733	43,29,438	5,21,01,005
15	5,00,188	62,01,734	2,10,000	4,232	15,06,934	44,84,799	5,65,85,804
16	4,97,687	64,17,554	2,10,000	2,539	15,61,678	46,45,876	6,12,31,680
17	4,95,199	66,40,885	2,10,000	1,523	16,18,142	48,12,743	6,60,44,423
18	4,92,723	68,71,988	2,10,000	914	16,76,459	49,85,529	7,10,29,952
19	4,90,259	71,11,133	2,10,000	548	17,36,739	51,64,394	7,61,94,346
20	4,87,808	73,58,600	2,10,000	329	17,99,077	53,49,523	8,15,43,869
21	4,85,369	76,14,679	2,10,000	197	18,63,560	55,41,119	8,70,84,988
22	4,82,942	78,79,670	2,10,000	118	19,30,273	57,39,397	9,28,24,385
23	4,80,527	81,53,883	2,10,000	71	19,99,299	59,44,584	9,87,68,969
24	4,78,125	84,37,638	2,10,000	43	20,70,721	61,56,917	10,49,25,886
25	4,75,734	87,31,268	2,10,000	26	21,44,626	63,76,641	11,13,02,527
26	4,73,355	90,35,116	2,10,000	15	22,21,101	66,04,015	11,79,06,542
27	4,70,988	93,49,538	2,10,000	9	23,00,237	68,39,301	12,47,45,843
28	4,68,634	96,74,902	2,10,000	6	23,82,125	70,82,777	13,18,28,620
29	4,66,290	1,00,11,588	2,10,000	3	24,66,863	73,34,725	13,91,63,345
30	4,63,959	1,03,59,992	2,10,000	2	25,54,549	75,95,442	14,67,58,787
Total	1,49,82,174	19,76,67,281	60,90,000	1,34,99,996	4,48,18,490	14,67,58,787	14,67,58,787

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

350 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.7 yrs
Discounted payback	4.9 yrs
Project IRR, unlevered	27.5%
Net present value at 10%	₹2.49 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.15 / kWh
Cumulative return, 30 years	987%
Total net cash flow, after tax	₹14.68 Cr
30-year generation	1,49,82,172 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.15
Grid energy charge today		₹7.65

Every unit the plant generates costs **₹3.15** across its life against **₹7.65** bought from APSPDCL today — and the grid figure reaches **₹23.86** by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 1,49,82,172 units, avoiding 12,285.4 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹1.42 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 350 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

APSPDCL · COMMERCIAL ROOFTOP

400 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

Net investment ₹1.54 Cr after input tax credit, generating 6,13,200 units in year one against an energy charge of ₹7.65 per unit.

30-YEAR ANALYSIS

PAYBACK

3.7 yrs

Post-tax, undiscounted

PROJECT IRR

27.5%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.15

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹2.85 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	400 kWp
Year 1 generation	6,13,200 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹43.91 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹40.50 / Wp (incl. GST)
Project price incl. GST	₹1.62 Cr
GST at 5% — recovered as ITC	₹7.71 L
Net capitalised cost	₹1.54 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹38.83 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,54,28,572	61,71,429	92,57,143	15,53,225
2	92,57,143	37,02,857	55,54,286	9,31,935
3	55,54,285	22,21,714	33,32,571	5,59,161
4	33,32,572	13,33,029	19,99,543	3,35,497
5	19,99,543	7,99,817	11,99,726	2,01,298
6	11,99,725	4,79,890	7,19,835	1,20,779
7	7,19,835	2,87,934	4,31,901	72,467
8	4,31,902	1,72,761	2,59,141	43,480
9	2,59,140	1,03,656	1,55,484	26,088
10	1,55,485	62,194	93,291	15,653
11	93,290	37,316	55,974	9,392
12	55,975	22,390	33,585	5,635
Tax shield, years 1-12				38,74,610

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

400 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	6,13,200	43,90,512	—	61,71,429	(4,48,221)	48,38,733	48,38,733
2	6,10,134	45,43,302	2,40,000	37,02,857	1,51,120	41,52,182	89,90,915
3	6,07,083	47,01,409	2,40,000	22,21,714	5,63,686	38,97,722	1,28,88,637
4	6,04,048	48,65,018	2,40,000	13,33,029	8,28,528	37,96,490	1,66,85,127
5	6,01,028	50,34,320	2,40,000	7,99,817	10,05,337	37,88,984	2,04,74,111
6	5,98,023	52,09,515	2,40,000	4,79,890	11,29,949	38,39,566	2,43,13,677
7	5,95,032	53,90,806	2,40,000	2,87,934	12,23,888	39,26,918	2,82,40,595
8	5,92,057	55,78,406	2,40,000	1,72,761	13,00,090	40,38,316	3,22,78,911
9	5,89,097	57,72,534	2,40,000	1,03,656	13,66,340	41,66,194	3,64,45,105
10	5,86,151	59,73,419	2,40,000	62,194	14,27,334	43,06,085	4,07,51,190
11	5,83,221	61,81,294	2,40,000	37,316	14,85,913	44,55,381	4,52,06,571
12	5,80,305	63,96,403	2,40,000	22,390	15,43,808	46,12,594	4,98,19,165
13	5,77,403	66,18,997	2,40,000	13,434	16,02,085	47,76,912	5,45,96,077
14	5,74,516	68,49,338	2,40,000	8,060	16,61,410	49,47,929	5,95,44,006
15	5,71,644	70,87,695	2,40,000	4,836	17,22,211	51,25,485	6,46,69,491
16	5,68,785	73,34,347	2,40,000	2,902	17,84,775	53,09,572	6,99,79,063
17	5,65,941	75,89,583	2,40,000	1,741	18,49,305	55,00,278	7,54,79,341
18	5,63,112	78,53,700	2,40,000	1,045	19,15,953	56,97,747	8,11,77,088
19	5,60,296	81,27,009	2,40,000	627	19,84,845	59,02,164	8,70,79,252
20	5,57,495	84,09,829	2,40,000	376	20,56,088	61,13,741	9,31,92,993
21	5,54,707	87,02,491	2,40,000	226	21,29,783	63,32,708	9,95,25,701
22	5,51,934	90,05,337	2,40,000	135	22,06,026	65,59,311	10,60,85,012
23	5,49,174	93,18,723	2,40,000	81	22,84,913	67,93,811	11,28,78,823
24	5,46,428	96,43,015	2,40,000	49	23,66,538	70,36,476	11,99,15,299
25	5,43,696	99,78,592	2,40,000	29	24,51,001	72,87,590	12,72,02,889
26	5,40,977	1,03,25,847	2,40,000	18	25,38,401	75,47,445	13,47,50,334
27	5,38,273	1,06,85,186	2,40,000	11	26,28,842	78,16,344	14,25,66,678
28	5,35,581	1,10,57,031	2,40,000	6	27,22,429	80,94,602	15,06,61,280
29	5,32,903	1,14,41,815	2,40,000	4	28,19,272	83,82,543	15,90,43,823
30	5,30,239	1,18,39,990	2,40,000	2	29,19,485	86,80,505	16,77,24,328
Total	1,71,22,483	22,59,05,463	69,60,000	1,54,28,569	5,12,21,134	16,77,24,328	16,77,24,328

RETURN METRICS

Simple payback	3.7 yrs
Discounted payback	4.9 yrs
Project IRR, unlevered	27.5%
Net present value at 10%	₹2.85 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.15 / kWh
Cumulative return, 30 years	987%
Total net cash flow, after tax	₹16.77 Cr
30-year generation	1,71,22,483 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.15
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.15 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 1,71,22,483 units, avoiding 14,040.4 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹1.62 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 400 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

APSPDCL · COMMERCIAL ROOFTOP

450 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

Net investment ₹1.74 Cr after input tax credit, generating 6,89,850 units in year one against an energy charge of ₹7.65 per unit.

30-YEAR ANALYSIS

PAYBACK

3.7 yrs

Post-tax, undiscounted

PROJECT IRR

27.5%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.15

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹3.20 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	450 kWp
Year 1 generation	6,89,850 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹49.39 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹40.50 / Wp (incl. GST)
Project price incl. GST	₹1.82 Cr
GST at 5% — recovered as ITC	₹8.68 L
Net capitalised cost	₹1.74 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹43.68 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,73,57,143	69,42,857	1,04,14,286	17,47,378
2	1,04,14,285	41,65,714	62,48,571	10,48,427
3	62,48,572	24,99,429	37,49,143	6,29,056
4	37,49,143	14,99,657	22,49,486	3,77,434
5	22,49,485	8,99,794	13,49,691	2,26,460
6	13,49,692	5,39,877	8,09,815	1,35,876
7	8,09,815	3,23,926	4,85,889	81,526
8	4,85,889	1,94,356	2,91,533	48,916
9	2,91,533	1,16,613	1,74,920	29,349
10	1,74,920	69,968	1,04,952	17,610
11	1,04,952	41,981	62,971	10,566
12	62,971	25,188	37,783	6,339
Tax shield, years 1-12				43,58,937

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

450 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	6,89,850	49,39,326	—	69,42,857	(5,04,249)	54,43,575	54,43,575
2	6,86,401	51,11,215	2,70,000	41,65,714	1,70,010	46,71,205	1,01,14,780
3	6,82,969	52,89,085	2,70,000	24,99,429	6,34,147	43,84,938	1,44,99,718
4	6,79,554	54,73,145	2,70,000	14,99,657	9,32,094	42,71,051	1,87,70,769
5	6,76,156	56,63,610	2,70,000	8,99,794	11,31,004	42,62,607	2,30,33,376
6	6,72,775	58,60,704	2,70,000	5,39,877	12,71,192	43,19,512	2,73,52,888
7	6,69,411	60,64,657	2,70,000	3,23,926	13,76,873	44,17,783	3,17,70,671
8	6,66,064	62,75,707	2,70,000	1,94,356	14,62,601	45,43,106	3,63,13,777
9	6,62,734	64,94,101	2,70,000	1,16,613	15,37,133	46,86,969	4,10,00,746
10	6,59,420	67,20,096	2,70,000	69,968	16,05,751	48,44,345	4,58,45,091
11	6,56,123	69,53,955	2,70,000	41,981	16,71,652	50,12,303	5,08,57,394
12	6,52,843	71,95,953	2,70,000	25,188	17,36,784	51,89,169	5,60,46,563
13	6,49,578	74,46,372	2,70,000	15,113	18,02,346	53,74,026	6,14,20,589
14	6,46,331	77,05,506	2,70,000	9,068	18,69,086	55,66,420	6,69,87,009
15	6,43,099	79,73,657	2,70,000	5,441	19,37,487	57,66,170	7,27,53,179
16	6,39,883	82,51,141	2,70,000	3,264	20,07,872	59,73,269	7,87,26,448
17	6,36,684	85,38,280	2,70,000	1,959	20,80,468	61,87,813	8,49,14,261
18	6,33,501	88,35,413	2,70,000	1,175	21,55,447	64,09,965	9,13,24,226
19	6,30,333	91,42,885	2,70,000	705	22,32,950	66,39,935	9,79,64,161
20	6,27,181	94,61,057	2,70,000	423	23,13,099	68,77,958	10,48,42,119
21	6,24,046	97,90,302	2,70,000	254	23,96,006	71,24,296	11,19,66,415
22	6,20,925	1,01,31,005	2,70,000	152	24,81,779	73,79,225	11,93,45,640
23	6,17,821	1,04,83,564	2,70,000	91	25,70,527	76,43,037	12,69,88,677
24	6,14,732	1,08,48,392	2,70,000	55	26,62,356	79,16,036	13,49,04,713
25	6,11,658	1,12,25,916	2,70,000	33	27,57,377	81,98,539	14,31,03,252
26	6,08,600	1,16,16,577	2,70,000	20	28,55,702	84,90,876	15,15,94,128
27	6,05,557	1,20,20,834	2,70,000	12	29,57,447	87,93,387	16,03,87,515
28	6,02,529	1,24,39,159	2,70,000	7	30,62,732	91,06,427	16,94,93,942
29	5,99,516	1,28,72,042	2,70,000	4	31,71,681	94,30,361	17,89,24,303
30	5,96,519	1,33,19,989	2,70,000	3	32,84,421	97,65,569	18,86,89,872
Total	1,92,62,793	25,41,43,645	78,30,000	1,73,57,139	5,76,23,775	18,86,89,872	18,86,89,872

RETURN METRICS

Simple payback	3.7 yrs
Discounted payback	4.9 yrs
Project IRR, unlevered	27.5%
Net present value at 10%	₹3.20 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.15 / kWh
Cumulative return, 30 years	987%
Total net cash flow, after tax	₹18.87 Cr
30-year generation	1,92,62,793 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.15
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.15 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 1,92,62,793 units, avoiding 15,795.5 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹1.82 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 450 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

APSPDCL · COMMERCIAL ROOFTOP

500 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

Net investment ₹1.93 Cr after input tax credit, generating 7,66,500 units in year one against an energy charge of ₹7.65 per unit.

30-YEAR ANALYSIS

PAYBACK

3.7 yrs

Post-tax, undiscounted

PROJECT IRR

27.5%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.15

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹3.56 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	500 kWp
Year 1 generation	7,66,500 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹54.88 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹40.50 / Wp (incl. GST)
Project price incl. GST	₹2.02 Cr
GST at 5% — recovered as ITC	₹9.64 L
Net capitalised cost	₹1.93 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹48.54 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,92,85,715	77,14,286	1,15,71,429	19,41,532
2	1,15,71,428	46,28,571	69,42,857	11,64,919
3	69,42,857	27,77,143	41,65,714	6,98,951
4	41,65,715	16,66,286	24,99,429	4,19,371
5	24,99,428	9,99,771	14,99,657	2,51,622
6	14,99,657	5,99,863	8,99,794	1,50,974
7	8,99,795	3,59,918	5,39,877	90,584
8	5,39,877	2,15,951	3,23,926	54,351
9	3,23,926	1,29,570	1,94,356	32,610
10	1,94,355	77,742	1,16,613	19,566
11	1,16,613	46,645	69,968	11,740
12	69,968	27,987	41,981	7,044
Tax shield, years 1-12				48,43,263

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

500 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	7,66,500	54,88,140	—	77,14,286	(5,60,276)	60,48,416	60,48,416
2	7,62,668	56,79,127	3,00,000	46,28,571	1,88,900	51,90,227	1,12,38,643
3	7,58,854	58,76,761	3,00,000	27,77,143	7,04,608	48,72,153	1,61,10,796
4	7,55,060	60,81,272	3,00,000	16,66,286	10,35,660	47,45,612	2,08,56,408
5	7,51,285	62,92,900	3,00,000	9,99,771	12,56,671	47,36,230	2,55,92,638
6	7,47,528	65,11,893	3,00,000	5,99,863	14,12,436	47,99,458	3,03,92,096
7	7,43,791	67,38,507	3,00,000	3,59,918	15,29,859	49,08,648	3,53,00,744
8	7,40,072	69,73,007	3,00,000	2,15,951	16,25,112	50,47,895	4,03,48,639
9	7,36,371	72,15,668	3,00,000	1,29,570	17,07,925	52,07,743	4,55,56,382
10	7,32,689	74,66,773	3,00,000	77,742	17,84,167	53,82,606	5,09,38,988
11	7,29,026	77,26,617	3,00,000	46,645	18,57,391	55,69,226	5,65,08,214
12	7,25,381	79,95,503	3,00,000	27,987	19,29,760	57,65,743	6,22,73,957
13	7,21,754	82,73,747	3,00,000	16,792	20,02,606	59,71,140	6,82,45,097
14	7,18,145	85,61,673	3,00,000	10,075	20,76,762	61,84,911	7,44,30,008
15	7,14,554	88,59,619	3,00,000	6,045	21,52,764	64,06,856	8,08,36,864
16	7,10,982	91,67,934	3,00,000	3,627	22,30,969	66,36,965	8,74,73,829
17	7,07,427	94,86,978	3,00,000	2,176	23,11,631	68,75,347	9,43,49,176
18	7,03,890	98,17,125	3,00,000	1,306	23,94,941	71,22,184	10,14,71,360
19	7,00,370	1,01,58,761	3,00,000	783	24,81,056	73,77,705	10,88,49,065
20	6,96,868	1,05,12,286	3,00,000	470	25,70,110	76,42,176	11,64,91,241
21	6,93,384	1,08,78,113	3,00,000	282	26,62,229	79,15,885	12,44,07,126
22	6,89,917	1,12,56,672	3,00,000	169	27,57,533	81,99,139	13,26,06,265
23	6,86,467	1,16,48,404	3,00,000	102	28,56,141	84,92,263	14,10,98,528
24	6,83,035	1,20,53,768	3,00,000	61	29,58,173	87,95,595	14,98,94,123
25	6,79,620	1,24,73,240	3,00,000	37	30,63,752	91,09,488	15,90,03,611
26	6,76,222	1,29,07,308	3,00,000	22	31,73,002	94,34,306	16,84,37,917
27	6,72,841	1,33,56,483	3,00,000	13	32,86,052	97,70,430	17,82,08,347
28	6,69,477	1,38,21,288	3,00,000	8	34,03,036	1,01,18,252	18,83,26,599
29	6,66,129	1,43,02,269	3,00,000	5	35,24,090	1,04,78,179	19,88,04,778
30	6,62,798	1,47,99,988	3,00,000	3	36,49,356	1,08,50,632	20,96,55,410
Total	2,14,03,105	28,23,81,824	87,00,000	1,92,85,709	6,40,26,416	20,96,55,410	20,96,55,410

RETURN METRICS

Simple payback	3.7 yrs
Discounted payback	4.9 yrs
Project IRR, unlevered	27.5%
Net present value at 10%	₹3.56 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.15 / kWh
Cumulative return, 30 years	987%
Total net cash flow, after tax	₹20.97 Cr
30-year generation	2,14,03,103 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.15
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.15 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,14,03,103 units, avoiding 17,550.5 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.02 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 500 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

550 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

550 KWP GRID-TIED SOLAR PLANT

Net investment ₹1.99 Cr after input tax credit, generating 8,43,150 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹4.02 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	550 kWp
Year 1 generation	8,43,150 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹60.37 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹2.09 Cr
GST at 5% — recovered as ITC	₹9.95 L
Net capitalised cost	₹1.99 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹50.10 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,99,04,762	79,61,905	1,19,42,857	20,03,852
2	1,19,42,857	47,77,143	71,65,714	12,02,311
3	71,65,715	28,66,286	42,99,429	7,21,387
4	42,99,428	17,19,771	25,79,657	4,32,832
5	25,79,657	10,31,863	15,47,794	2,59,699
6	15,47,795	6,19,118	9,28,677	1,55,820
7	9,28,677	3,71,471	5,57,206	93,492
8	5,57,206	2,22,882	3,34,324	56,095
9	3,34,323	1,33,729	2,00,594	33,657
10	2,00,594	80,238	1,20,356	20,194
11	1,20,357	48,143	72,214	12,117
12	72,214	28,886	43,328	7,270
Tax shield, years 1-12				49,98,726

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

550 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	8,43,150	60,36,954	—	79,61,905	(4,84,472)	65,21,426	65,21,426
2	8,38,934	62,47,040	3,30,000	47,77,143	2,86,889	56,30,151	1,21,51,577
3	8,34,740	64,64,437	3,30,000	28,66,286	8,22,528	53,11,909	1,74,63,486
4	8,30,566	66,89,399	3,30,000	17,19,771	11,67,702	51,91,698	2,26,55,184
5	8,26,413	69,22,190	3,30,000	10,31,863	13,99,423	51,92,767	2,78,47,951
6	8,22,281	71,63,083	3,30,000	6,19,118	15,63,931	52,69,152	3,31,17,103
7	8,18,170	74,12,358	3,30,000	3,71,471	16,88,996	53,93,362	3,85,10,465
8	8,14,079	76,70,308	3,30,000	2,22,882	17,91,314	55,48,994	4,40,59,459
9	8,10,008	79,37,235	3,30,000	1,33,729	18,80,932	57,26,303	4,97,85,762
10	8,05,958	82,13,451	3,30,000	80,238	19,63,913	59,19,538	5,57,05,300
11	8,01,929	84,99,279	3,30,000	48,143	20,43,928	61,25,351	6,18,30,651
12	7,97,919	87,95,054	3,30,000	28,886	21,23,215	63,41,839	6,81,72,490
13	7,93,929	91,01,121	3,30,000	17,331	22,03,154	65,67,968	7,47,40,458
14	7,89,960	94,17,840	3,30,000	10,399	22,84,611	68,03,230	8,15,43,688
15	7,86,010	97,45,581	3,30,000	6,239	23,68,143	70,47,438	8,85,91,126
16	7,82,080	1,00,84,727	3,30,000	3,744	24,54,128	73,00,600	9,58,91,726
17	7,78,169	1,04,35,676	3,30,000	2,246	25,42,831	75,62,845	10,34,54,571
18	7,74,279	1,07,98,838	3,30,000	1,348	26,34,458	78,34,380	11,12,88,951
19	7,70,407	1,11,74,637	3,30,000	809	27,29,175	81,15,462	11,94,04,413
20	7,66,555	1,15,63,514	3,30,000	485	28,27,129	84,06,386	12,78,10,799
21	7,62,722	1,19,65,925	3,30,000	291	29,28,456	87,07,468	13,65,18,267
22	7,58,909	1,23,82,339	3,30,000	175	30,33,289	90,19,050	14,55,37,317
23	7,55,114	1,28,13,244	3,30,000	105	31,41,757	93,41,488	15,48,78,805
24	7,51,339	1,32,59,145	3,30,000	63	32,53,991	96,75,154	16,45,53,959
25	7,47,582	1,37,20,563	3,30,000	38	33,70,128	1,00,20,436	17,45,74,395
26	7,43,844	1,41,98,039	3,30,000	23	34,90,302	1,03,77,737	18,49,52,132
27	7,40,125	1,46,92,131	3,30,000	14	36,14,658	1,07,47,473	19,56,99,605
28	7,36,424	1,52,03,417	3,30,000	8	37,43,340	1,11,30,077	20,68,29,682
29	7,32,742	1,57,32,496	3,30,000	5	38,76,499	1,15,25,997	21,83,55,679
30	7,29,078	1,62,79,987	3,30,000	3	40,14,292	1,19,35,695	23,02,91,374
Total	2,35,43,415	31,06,20,008	95,70,000	1,99,04,761	7,07,58,640	23,02,91,374	23,02,91,374

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

550 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹4.02 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹23.03 Cr
30-year generation	2,35,43,414 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,35,43,414 units, avoiding 19,305.6 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.09 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 550 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

APSPDCL · COMMERCIAL ROOFTOP

600 KWP GRID-TIED SOLAR PLANT

Net investment ₹2.17 Cr after input tax credit, generating 9,19,800 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹4.39 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	600 kWp
Year 1 generation	9,19,800 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹65.86 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹2.28 Cr
GST at 5% — recovered as ITC	₹10.86 L
Net capitalised cost	₹2.17 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹54.65 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	2,17,14,285	86,85,714	1,30,28,571	21,86,020
2	1,30,28,572	52,11,429	78,17,143	13,11,612
3	78,17,143	31,26,857	46,90,286	7,86,967
4	46,90,285	18,76,114	28,14,171	4,72,180
5	28,14,172	11,25,669	16,88,503	2,83,308
6	16,88,503	6,75,401	10,13,102	1,69,985
7	10,13,102	4,05,241	6,07,861	1,01,991
8	6,07,861	2,43,144	3,64,717	61,194
9	3,64,717	1,45,887	2,18,830	36,717
10	2,18,830	87,532	1,31,298	22,030
11	1,31,298	52,519	78,779	13,218
12	78,779	31,512	47,267	7,931
Tax shield, years 1-12				54,53,155

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

600 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	9,19,800	65,85,768	—	86,85,714	(5,28,514)	71,14,282	71,14,282
2	9,15,201	68,14,953	3,60,000	52,11,429	3,12,970	61,41,983	1,32,56,265
3	9,10,625	70,52,113	3,60,000	31,26,857	8,97,304	57,94,809	1,90,51,074
4	9,06,072	72,97,527	3,60,000	18,76,114	12,73,856	56,63,670	2,47,14,744
5	9,01,542	75,51,481	3,60,000	11,25,669	15,26,644	56,64,837	3,03,79,581
6	8,97,034	78,14,272	3,60,000	6,75,401	17,06,106	57,48,166	3,61,27,747
7	8,92,549	80,86,209	3,60,000	4,05,241	18,42,541	58,83,667	4,20,11,414
8	8,88,086	83,67,609	3,60,000	2,43,144	19,54,160	60,53,448	4,80,64,862
9	8,83,645	86,58,802	3,60,000	1,45,887	20,51,926	62,46,876	5,43,11,738
10	8,79,227	89,60,128	3,60,000	87,532	21,42,450	64,57,678	6,07,69,416
11	8,74,831	92,71,940	3,60,000	52,519	22,29,739	66,82,201	6,74,51,617
12	8,70,457	95,94,604	3,60,000	31,512	23,16,234	69,18,370	7,43,69,987
13	8,66,105	99,28,496	3,60,000	18,907	24,03,441	71,65,055	8,15,35,042
14	8,61,774	1,02,74,008	3,60,000	11,344	24,92,302	74,21,705	8,89,56,747
15	8,57,465	1,06,31,543	3,60,000	6,806	25,83,429	76,88,114	9,66,44,861
16	8,53,178	1,10,01,521	3,60,000	4,084	26,77,230	79,64,291	10,46,09,152
17	8,48,912	1,13,84,374	3,60,000	2,450	27,73,998	82,50,376	11,28,59,528
18	8,44,667	1,17,80,550	3,60,000	1,470	28,73,954	85,46,596	12,14,06,124
19	8,40,444	1,21,90,513	3,60,000	882	29,77,282	88,53,232	13,02,59,356
20	8,36,242	1,26,14,743	3,60,000	529	30,84,141	91,70,603	13,94,29,959
21	8,32,061	1,30,53,736	3,60,000	318	31,94,680	94,99,057	14,89,29,016
22	8,27,900	1,35,08,006	3,60,000	191	33,09,042	98,38,964	15,87,67,980
23	8,23,761	1,39,78,085	3,60,000	114	34,27,371	1,01,90,714	16,89,58,694
24	8,19,642	1,44,64,522	3,60,000	69	35,49,809	1,05,54,713	17,95,13,407
25	8,15,544	1,49,67,887	3,60,000	41	36,76,503	1,09,31,385	19,04,44,792
26	8,11,466	1,54,88,770	3,60,000	25	38,07,603	1,13,21,167	20,17,65,959
27	8,07,409	1,60,27,779	3,60,000	15	39,43,263	1,17,24,516	21,34,90,475
28	8,03,372	1,65,85,546	3,60,000	9	40,83,643	1,21,41,903	22,56,32,378
29	7,99,355	1,71,62,723	3,60,000	5	42,28,908	1,25,73,815	23,82,06,193
30	7,95,358	1,77,59,986	3,60,000	3	43,79,228	1,30,20,758	25,12,26,951
Total	2,56,83,724	33,88,58,194	1,04,40,000	2,17,14,281	7,71,91,243	25,12,26,951	25,12,26,951

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

600 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹4.39 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹25.12 Cr
30-year generation	2,56,83,724 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,56,83,724 units, avoiding 21,060.7 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.28 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 600 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

650 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

650 KWP GRID-TIED SOLAR PLANT

Net investment ₹2.35 Cr after input tax credit, generating 9,96,450 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹4.75 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	650 kWp
Year 1 generation	9,96,450 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹71.35 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹2.47 Cr
GST at 5% — recovered as ITC	₹11.76 L
Net capitalised cost	₹2.35 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹59.20 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	2,35,23,810	94,09,524	1,41,14,286	23,68,189
2	1,41,14,285	56,45,714	84,68,571	14,20,913
3	84,68,572	33,87,429	50,81,143	8,52,548
4	50,81,143	20,32,457	30,48,686	5,11,529
5	30,48,685	12,19,474	18,29,211	3,06,917
6	18,29,212	7,31,685	10,97,527	1,84,150
7	10,97,527	4,39,011	6,58,516	1,10,490
8	6,58,516	2,63,406	3,95,110	66,294
9	3,95,110	1,58,044	2,37,066	39,777
10	2,37,065	94,826	1,42,239	23,866
11	1,42,240	56,896	85,344	14,320
12	85,343	34,137	51,206	8,592
Tax shield, years 1-12				59,07,585

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

650 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	9,96,450	71,34,582	—	94,09,524	(5,72,557)	77,07,139	77,07,139
2	9,91,468	73,82,865	3,90,000	56,45,714	3,39,051	66,53,814	1,43,60,953
3	9,86,510	76,39,789	3,90,000	33,87,429	9,72,079	62,77,710	2,06,38,663
4	9,81,578	79,05,654	3,90,000	20,32,457	13,80,011	61,35,643	2,67,74,306
5	9,76,670	81,80,771	3,90,000	12,19,474	16,53,864	61,36,907	3,29,11,213
6	9,71,787	84,65,461	3,90,000	7,31,685	18,48,282	62,27,180	3,91,38,393
7	9,66,928	87,60,059	3,90,000	4,39,011	19,96,086	63,73,973	4,55,12,366
8	9,62,093	90,64,910	3,90,000	2,63,406	21,17,007	65,57,902	5,20,70,268
9	9,57,283	93,80,368	3,90,000	1,58,044	22,22,919	67,67,449	5,88,37,717
10	9,52,496	97,06,805	3,90,000	94,826	23,20,988	69,95,818	6,58,33,535
11	9,47,734	1,00,44,602	3,90,000	56,896	24,15,551	72,39,051	7,30,72,586
12	9,42,995	1,03,94,154	3,90,000	34,137	25,09,254	74,94,900	8,05,67,486
13	9,38,280	1,07,55,871	3,90,000	20,482	26,03,727	77,62,143	8,83,29,629
14	9,33,589	1,11,30,175	3,90,000	12,289	26,99,994	80,40,181	9,63,69,810
15	9,28,921	1,15,17,505	3,90,000	7,374	27,98,715	83,28,790	10,46,98,600
16	9,24,276	1,19,18,314	3,90,000	4,424	29,00,333	86,27,982	11,33,26,582
17	9,19,655	1,23,33,072	3,90,000	2,655	30,05,164	89,37,907	12,22,64,489
18	9,15,056	1,27,62,263	3,90,000	1,593	31,13,450	92,58,812	13,15,23,301
19	9,10,481	1,32,06,389	3,90,000	956	32,25,388	95,91,001	14,11,14,302
20	9,05,929	1,36,65,972	3,90,000	573	33,41,152	99,34,819	15,10,49,121
21	9,01,399	1,41,41,547	3,90,000	344	34,60,903	1,02,90,645	16,13,39,766
22	8,96,892	1,46,33,673	3,90,000	206	35,84,796	1,06,58,878	17,19,98,644
23	8,92,408	1,51,42,925	3,90,000	124	37,12,985	1,10,39,940	18,30,38,584
24	8,87,946	1,56,69,899	3,90,000	74	38,45,626	1,14,34,273	19,44,72,857
25	8,83,506	1,62,15,211	3,90,000	45	39,82,878	1,18,42,333	20,63,15,190
26	8,79,088	1,67,79,501	3,90,000	27	41,24,903	1,22,64,598	21,85,79,788
27	8,74,693	1,73,63,427	3,90,000	16	42,71,868	1,27,01,559	23,12,81,347
28	8,70,319	1,79,67,675	3,90,000	10	44,23,947	1,31,53,728	24,44,35,075
29	8,65,968	1,85,92,950	3,90,000	6	45,81,317	1,36,21,633	25,80,56,708
30	8,61,638	1,92,39,984	3,90,000	3	47,44,163	1,41,05,821	27,21,62,529
Total	2,78,24,036	36,70,96,373	1,13,10,000	2,35,23,804	8,36,23,844	27,21,62,529	27,21,62,529

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹4.75 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹27.22 Cr
30-year generation	2,78,24,034 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,78,24,034 units, avoiding 22,815.7 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.47 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 650 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

APSPDCL · COMMERCIAL ROOFTOP

700 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

Net investment ₹2.53 Cr after input tax credit, generating 10,73,100 units in year one against an energy charge of ₹7.65 per unit.

PAYBACK 3.5 yrs Post-tax, undiscounted	PROJECT IRR 29.0% Unlevered, 30 years	COST OF SOLAR ENERGY ₹2.98 per kWh, against ₹7.65 from the grid	NET PRESENT VALUE ₹5.12 Cr Discounted at 10%
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PLANT AND GENERATION	INVESTMENT AND TAX BASIS																																
<table><tr><td>Plant capacity</td><td>700 kWp</td></tr><tr><td>Year 1 generation</td><td>10,73,100 kWh</td></tr><tr><td>Specific yield</td><td>1,533 kWh / kWp / year</td></tr><tr><td>Supply category</td><td>HT Category I / II at 11 kV (HT supply at 11 kV)</td></tr><tr><td>Energy charge offset</td><td>₹7.65 / kWh</td></tr><tr><td>Units self-consumed</td><td>90% at the energy charge</td></tr><tr><td>Units exported</td><td>10% at ₹2.75 / kWh (APPC)</td></tr><tr><td>Year 1 gross saving</td><td>₹76.83 L</td></tr></table>	Plant capacity	700 kWp	Year 1 generation	10,73,100 kWh	Specific yield	1,533 kWh / kWp / year	Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)	Energy charge offset	₹7.65 / kWh	Units self-consumed	90% at the energy charge	Units exported	10% at ₹2.75 / kWh (APPC)	Year 1 gross saving	₹76.83 L	<table><tr><td>Price per watt-peak</td><td>₹38.00 / Wp (incl. GST)</td></tr><tr><td>Project price incl. GST</td><td>₹2.66 Cr</td></tr><tr><td>GST at 5% — recovered as ITC</td><td>₹12.67 L</td></tr><tr><td>Net capitalised cost</td><td>₹2.53 Cr</td></tr><tr><td>Depreciation block</td><td>40% WDV, Appendix I</td></tr><tr><td>Tax regime</td><td>s.115BAA — 25.168%</td></tr><tr><td>Annual maintenance</td><td>₹600 / kWp / year, year 1 free</td></tr><tr><td>30-year depreciation shield</td><td>₹63.76 L</td></tr></table>	Price per watt-peak	₹38.00 / Wp (incl. GST)	Project price incl. GST	₹2.66 Cr	GST at 5% — recovered as ITC	₹12.67 L	Net capitalised cost	₹2.53 Cr	Depreciation block	40% WDV, Appendix I	Tax regime	s.115BAA — 25.168%	Annual maintenance	₹600 / kWp / year, year 1 free	30-year depreciation shield	₹63.76 L
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30-year depreciation shield	₹63.76 L																																

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	2,53,33,333	1,01,33,333	1,52,00,000	25,50,357
2	1,52,00,000	60,80,000	91,20,000	15,30,214
3	91,20,000	36,48,000	54,72,000	9,18,129
4	54,72,000	21,88,800	32,83,200	5,50,877
5	32,83,200	13,13,280	19,69,920	3,30,526
6	19,69,920	7,87,968	11,81,952	1,98,316
7	11,81,952	4,72,781	7,09,171	1,18,990
8	7,09,171	2,83,668	4,25,503	71,394
9	4,25,503	1,70,201	2,55,302	42,836
10	2,55,302	1,02,121	1,53,181	25,702
11	1,53,181	61,272	91,909	15,421
12	91,908	36,763	55,145	9,253
Tax shield, years 1-12				63,62,014

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

700 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	10,73,100	76,83,396	—	1,01,33,333	(6,16,600)	82,99,996	82,99,996
2	10,67,735	79,50,778	4,20,000	60,80,000	3,65,132	71,65,646	1,54,65,642
3	10,62,396	82,27,465	4,20,000	36,48,000	10,46,854	67,60,611	2,22,26,253
4	10,57,084	85,13,781	4,20,000	21,88,800	14,86,166	66,07,615	2,88,33,868
5	10,51,798	88,10,061	4,20,000	13,13,280	17,81,084	66,08,976	3,54,42,844
6	10,46,539	91,16,651	4,20,000	7,87,968	19,90,457	67,06,193	4,21,49,037
7	10,41,307	94,33,910	4,20,000	4,72,781	21,49,631	68,64,279	4,90,13,316
8	10,36,100	97,62,210	4,20,000	2,83,668	22,79,854	70,62,356	5,60,75,672
9	10,30,920	1,01,01,935	4,20,000	1,70,201	23,93,913	72,88,022	6,33,63,694
10	10,25,765	1,04,53,483	4,20,000	1,02,121	24,99,525	75,33,957	7,08,97,651
11	10,20,636	1,08,17,264	4,20,000	61,272	26,01,362	77,95,901	7,86,93,552
12	10,15,533	1,11,93,704	4,20,000	36,763	27,02,273	80,71,431	8,67,64,983
13	10,10,455	1,15,83,245	4,20,000	22,058	28,04,014	83,59,231	9,51,24,214
14	10,05,403	1,19,86,342	4,20,000	13,235	29,07,686	86,58,656	10,37,82,870
15	10,00,376	1,24,03,467	4,20,000	7,941	30,14,000	89,69,467	11,27,52,337
16	9,95,374	1,28,35,108	4,20,000	4,765	31,23,435	92,91,673	12,20,44,010
17	9,90,397	1,32,81,769	4,20,000	2,859	32,36,331	96,25,439	13,16,69,449
18	9,85,445	1,37,43,975	4,20,000	1,715	33,52,946	99,71,029	14,16,40,478
19	9,80,518	1,42,22,265	4,20,000	1,029	34,73,495	1,03,28,770	15,19,69,248
20	9,75,616	1,47,17,200	4,20,000	617	35,98,164	1,06,99,036	16,26,68,284
21	9,70,738	1,52,29,359	4,20,000	370	37,27,126	1,10,82,233	17,37,50,517
22	9,65,884	1,57,59,340	4,20,000	222	38,60,549	1,14,78,791	18,52,29,308
23	9,61,054	1,63,07,766	4,20,000	133	39,98,599	1,18,89,166	19,71,18,474
24	9,56,249	1,68,75,276	4,20,000	80	41,41,444	1,23,13,832	20,94,32,306
25	9,51,468	1,74,62,535	4,20,000	48	42,89,253	1,27,53,282	22,21,85,588
26	9,46,711	1,80,70,232	4,20,000	29	44,42,203	1,32,08,029	23,53,93,617
27	9,41,977	1,86,99,076	4,20,000	17	46,00,473	1,36,78,602	24,90,72,219
28	9,37,267	1,93,49,803	4,20,000	10	47,64,250	1,41,65,553	26,32,37,772
29	9,32,581	2,00,23,177	4,20,000	6	49,33,726	1,46,69,451	27,79,07,223
30	9,27,918	2,07,19,983	4,20,000	4	51,09,099	1,51,90,884	29,30,98,107
Total	2,99,64,344	39,53,34,556	1,21,80,000	2,53,33,325	9,00,56,444	29,30,98,107	29,30,98,107

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹5.12 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹29.31 Cr
30-year generation	2,99,64,345 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,99,64,345 units, avoiding 24,570.8 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.66 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 700 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

750 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

750 KWP GRID-TIED SOLAR PLANT

Net investment ₹2.71 Cr after input tax credit, generating 11,49,750 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹5.48 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	750 kWp
Year 1 generation	11,49,750 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹82.32 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹2.85 Cr
GST at 5% — recovered as ITC	₹13.57 L
Net capitalised cost	₹2.71 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹68.31 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	2,71,42,857	1,08,57,143	1,62,85,714	27,32,526
2	1,62,85,715	65,14,286	97,71,429	16,39,516
3	97,71,428	39,08,571	58,62,857	9,83,709
4	58,62,857	23,45,143	35,17,714	5,90,226
5	35,17,715	14,07,086	21,10,629	3,54,135
6	21,10,628	8,44,251	12,66,377	2,12,481
7	12,66,377	5,06,551	7,59,826	1,27,489
8	7,59,827	3,03,931	4,55,896	76,493
9	4,55,895	1,82,358	2,73,537	45,896
10	2,73,537	1,09,415	1,64,122	27,538
11	1,64,122	65,649	98,473	16,523
12	98,473	39,389	59,084	9,913
Tax shield, years 1-12				68,16,444

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

750 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	11,49,750	82,32,210	—	1,08,57,143	(6,60,643)	88,92,853	88,92,853
2	11,44,001	85,18,691	4,50,000	65,14,286	3,91,213	76,77,478	1,65,70,331
3	11,38,281	88,15,141	4,50,000	39,08,571	11,21,630	72,43,512	2,38,13,843
4	11,32,590	91,21,908	4,50,000	23,45,143	15,92,320	70,79,588	3,08,93,431
5	11,26,927	94,39,351	4,50,000	14,07,086	19,08,304	70,81,046	3,79,74,477
6	11,21,292	97,67,840	4,50,000	8,44,251	21,32,633	71,85,207	4,51,59,684
7	11,15,686	1,01,07,761	4,50,000	5,06,551	23,03,177	73,54,584	5,25,14,268
8	11,10,107	1,04,59,511	4,50,000	3,03,931	24,42,700	75,66,811	6,00,81,079
9	11,04,557	1,08,23,502	4,50,000	1,82,358	25,64,907	78,08,595	6,78,89,674
10	10,99,034	1,12,00,160	4,50,000	1,09,415	26,78,063	80,72,097	7,59,61,771
11	10,93,539	1,15,89,925	4,50,000	65,649	27,87,174	83,52,752	8,43,14,523
12	10,88,071	1,19,93,255	4,50,000	39,389	28,95,293	86,47,962	9,29,62,485
13	10,82,631	1,24,10,620	4,50,000	23,634	30,04,301	89,56,319	10,19,18,804
14	10,77,218	1,28,42,510	4,50,000	14,180	31,15,378	92,77,132	11,11,95,936
15	10,71,832	1,32,89,429	4,50,000	8,508	32,29,286	96,10,143	12,08,06,079
16	10,66,472	1,37,51,901	4,50,000	5,105	33,46,538	99,55,363	13,07,61,442
17	10,61,140	1,42,30,467	4,50,000	3,063	34,67,497	1,03,12,970	14,10,74,412
18	10,55,834	1,47,25,688	4,50,000	1,838	35,92,443	1,06,83,245	15,17,57,657
19	10,50,555	1,52,38,141	4,50,000	1,103	37,21,602	1,10,66,540	16,28,24,197
20	10,45,302	1,57,68,429	4,50,000	662	38,55,176	1,14,63,253	17,42,87,450
21	10,40,076	1,63,17,170	4,50,000	397	39,93,349	1,18,73,821	18,61,61,271
22	10,34,876	1,68,85,008	4,50,000	238	41,36,303	1,22,98,705	19,84,59,976
23	10,29,701	1,74,72,606	4,50,000	143	42,84,213	1,27,38,392	21,11,98,368
24	10,24,553	1,80,80,653	4,50,000	86	44,37,261	1,31,93,392	22,43,91,760
25	10,19,430	1,87,09,859	4,50,000	51	45,95,628	1,36,64,231	23,80,55,991
26	10,14,333	1,93,60,962	4,50,000	31	47,59,503	1,41,51,459	25,22,07,450
27	10,09,261	2,00,34,724	4,50,000	19	49,29,079	1,46,55,645	26,68,63,095
28	10,04,215	2,07,31,932	4,50,000	11	51,04,554	1,51,77,378	28,20,40,473
29	9,99,194	2,14,53,404	4,50,000	7	52,86,135	1,57,17,269	29,77,57,742
30	9,94,198	2,21,99,982	4,50,000	4	54,74,034	1,62,75,948	31,40,33,690
Total	3,21,04,656	42,35,72,740	1,30,50,000	2,71,42,853	9,64,89,051	31,40,33,690	31,40,33,690

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹5.48 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹31.40 Cr
30-year generation	3,21,04,655 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 3,21,04,655 units, avoiding 26,325.8 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.85 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 750 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

APSPDCL · COMMERCIAL ROOFTOP

800 KWP GRID-TIED SOLAR PLANT

Net investment ₹2.90 Cr after input tax credit, generating 12,26,400 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹5.85 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	800 kWp
Year 1 generation	12,26,400 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹87.81 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹3.04 Cr
GST at 5% — recovered as ITC	₹14.48 L
Net capitalised cost	₹2.90 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹72.87 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	2,89,52,381	1,15,80,952	1,73,71,429	29,14,694
2	1,73,71,428	69,48,571	1,04,22,857	17,48,816
3	1,04,22,857	41,69,143	62,53,714	10,49,290
4	62,53,715	25,01,486	37,52,229	6,29,574
5	37,52,228	15,00,891	22,51,337	3,77,744
6	22,51,337	9,00,535	13,50,802	2,26,647
7	13,50,802	5,40,321	8,10,481	1,35,988
8	8,10,482	3,24,193	4,86,289	81,593
9	4,86,289	1,94,516	2,91,773	48,956
10	2,91,773	1,16,709	1,75,064	29,373
11	1,75,064	70,026	1,05,038	17,624
12	1,05,038	42,015	63,023	10,574
Tax shield, years 1-12				72,70,874

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

800 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	12,26,400	87,81,024	—	1,15,80,952	(7,04,686)	94,85,710	94,85,710
2	12,20,268	90,86,604	4,80,000	69,48,571	4,17,294	81,89,310	1,76,75,020
3	12,14,167	94,02,817	4,80,000	41,69,143	11,96,405	77,26,413	2,54,01,433
4	12,08,096	97,30,035	4,80,000	25,01,486	16,98,475	75,51,560	3,29,52,993
5	12,02,055	1,00,68,641	4,80,000	15,00,891	20,35,525	75,53,116	4,05,06,109
6	11,96,045	1,04,19,029	4,80,000	9,00,535	22,74,808	76,64,221	4,81,70,330
7	11,90,065	1,07,81,612	4,80,000	5,40,321	24,56,722	78,44,890	5,60,15,220
8	11,84,115	1,11,56,812	4,80,000	3,24,193	26,05,547	80,71,265	6,40,86,485
9	11,78,194	1,15,45,069	4,80,000	1,94,516	27,35,901	83,29,168	7,24,15,653
10	11,72,303	1,19,46,837	4,80,000	1,16,709	28,56,600	86,10,237	8,10,25,890
11	11,66,441	1,23,62,587	4,80,000	70,026	29,72,985	89,09,602	8,99,35,492
12	11,60,609	1,27,92,805	4,80,000	42,015	30,88,312	92,24,493	9,91,59,985
13	11,54,806	1,32,37,995	4,80,000	25,209	32,04,587	95,53,407	10,87,13,392
14	11,49,032	1,36,98,677	4,80,000	15,126	33,23,070	98,95,607	11,86,08,999
15	11,43,287	1,41,75,391	4,80,000	9,075	34,44,572	1,02,50,819	12,88,59,818
16	11,37,571	1,46,68,695	4,80,000	5,445	35,69,640	1,06,19,054	13,94,78,872
17	11,31,883	1,51,79,165	4,80,000	3,267	36,98,664	1,10,00,501	15,04,79,373
18	11,26,223	1,57,07,400	4,80,000	1,960	38,31,939	1,13,95,461	16,18,74,834
19	11,20,592	1,62,54,018	4,80,000	1,176	39,69,709	1,18,04,309	17,36,79,143
20	11,14,989	1,68,19,657	4,80,000	706	41,12,187	1,22,27,470	18,59,06,613
21	11,09,414	1,74,04,981	4,80,000	423	42,59,573	1,26,65,409	19,85,72,022
22	11,03,867	1,80,10,675	4,80,000	254	44,12,056	1,31,18,619	21,16,90,641
23	10,98,348	1,86,37,446	4,80,000	152	45,69,828	1,35,87,619	22,52,78,260
24	10,92,856	1,92,86,029	4,80,000	91	47,33,078	1,40,72,951	23,93,51,211
25	10,87,392	1,99,57,183	4,80,000	55	49,02,004	1,45,75,180	25,39,26,391
26	10,81,955	2,06,51,693	4,80,000	33	50,76,803	1,50,94,890	26,90,21,281
27	10,76,545	2,13,70,372	4,80,000	20	52,57,684	1,56,32,688	28,46,53,969
28	10,71,162	2,21,14,061	4,80,000	12	54,44,858	1,61,89,204	30,08,43,173
29	10,65,807	2,28,83,630	4,80,000	7	56,38,544	1,67,65,087	31,76,08,260
30	10,60,478	2,36,79,981	4,80,000	4	58,38,970	1,73,61,011	33,49,69,271
Total	3,42,44,965	45,18,10,921	1,39,20,000	2,89,52,373	10,29,21,654	33,49,69,271	33,49,69,271

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹5.85 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹33.50 Cr
30-year generation	3,42,44,965 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 3,42,44,965 units, avoiding 28,080.9 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹3.04 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 800 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

850 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

850 KWP GRID-TIED SOLAR PLANT

Net investment ₹3.08 Cr after input tax credit, generating 13,03,050 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹6.21 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	850 kWp
Year 1 generation	13,03,050 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹93.30 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹3.23 Cr
GST at 5% — recovered as ITC	₹15.38 L
Net capitalised cost	₹3.08 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹77.42 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	3,07,61,905	1,23,04,762	1,84,57,143	30,96,863
2	1,84,57,143	73,82,857	1,10,74,286	18,58,117
3	1,10,74,285	44,29,714	66,44,571	11,14,870
4	66,44,572	26,57,829	39,86,743	6,68,922
5	39,86,743	15,94,697	23,92,046	4,01,353
6	23,92,045	9,56,818	14,35,227	2,40,812
7	14,35,227	5,74,091	8,61,136	1,44,487
8	8,61,137	3,44,455	5,16,682	86,692
9	5,16,682	2,06,673	3,10,009	52,015
10	3,10,009	1,24,004	1,86,005	31,209
11	1,86,005	74,402	1,11,603	18,725
12	1,11,603	44,641	66,962	11,235
Tax shield, years 1-12				77,25,303

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

850 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	13,03,050	93,29,838	—	1,23,04,762	(7,48,729)	1,00,78,567	1,00,78,567
2	12,96,535	96,54,516	5,10,000	73,82,857	4,43,374	87,01,142	1,87,79,709
3	12,90,052	99,90,494	5,10,000	44,29,714	12,71,180	82,09,313	2,69,89,022
4	12,83,602	1,03,38,163	5,10,000	26,57,829	18,04,630	80,23,533	3,50,12,555
5	12,77,184	1,06,97,931	5,10,000	15,94,697	21,62,745	80,25,186	4,30,37,741
6	12,70,798	1,10,70,219	5,10,000	9,56,818	24,16,984	81,43,235	5,11,80,976
7	12,64,444	1,14,55,462	5,10,000	5,74,091	26,10,267	83,35,196	5,95,16,172
8	12,58,122	1,18,54,112	5,10,000	3,44,455	27,68,394	85,75,719	6,80,91,891
9	12,51,831	1,22,66,636	5,10,000	2,06,673	29,06,895	88,49,741	7,69,41,632
10	12,45,572	1,26,93,514	5,10,000	1,24,004	30,35,138	91,48,377	8,60,90,009
11	12,39,344	1,31,35,249	5,10,000	74,402	31,58,797	94,66,452	9,55,56,461
12	12,33,147	1,35,92,355	5,10,000	44,641	32,81,332	98,01,024	10,53,57,485
13	12,26,982	1,40,65,369	5,10,000	26,785	34,04,874	1,01,50,495	11,55,07,980
14	12,20,847	1,45,54,844	5,10,000	16,071	35,30,762	1,05,14,083	12,60,22,063
15	12,14,742	1,50,61,353	5,10,000	9,643	36,59,858	1,08,91,495	13,69,13,558
16	12,08,669	1,55,85,488	5,10,000	5,786	37,92,743	1,12,82,745	14,81,96,303
17	12,02,625	1,61,27,863	5,10,000	3,471	39,29,830	1,16,88,033	15,98,84,336
18	11,96,612	1,66,89,113	5,10,000	2,083	40,71,435	1,21,07,678	17,19,92,014
19	11,90,629	1,72,69,894	5,10,000	1,250	42,17,816	1,25,42,078	18,45,34,092
20	11,84,676	1,78,70,886	5,10,000	750	43,69,199	1,29,91,687	19,75,25,779
21	11,78,753	1,84,92,793	5,10,000	450	45,25,796	1,34,56,997	21,09,82,776
22	11,72,859	1,91,36,342	5,10,000	270	46,87,810	1,39,38,532	22,49,21,308
23	11,66,995	1,98,02,287	5,10,000	162	48,55,442	1,44,36,845	23,93,58,153
24	11,61,160	2,04,91,406	5,10,000	97	50,28,896	1,49,52,510	25,43,10,663
25	11,55,354	2,12,04,507	5,10,000	58	52,08,379	1,54,86,128	26,97,96,791
26	11,49,577	2,19,42,424	5,10,000	35	53,94,104	1,60,38,320	28,58,35,111
27	11,43,829	2,27,06,020	5,10,000	21	55,86,289	1,66,09,731	30,24,44,842
28	11,38,110	2,34,96,190	5,10,000	13	57,85,161	1,72,01,029	31,96,45,871
29	11,32,420	2,43,13,857	5,10,000	8	59,90,953	1,78,12,904	33,74,58,775
30	11,26,757	2,51,59,980	5,10,000	5	62,03,906	1,84,46,074	35,59,04,849
Total	3,63,85,277	48,00,49,105	1,47,90,000	3,07,61,901	10,93,54,260	35,59,04,849	35,59,04,849

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹6.21 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹35.59 Cr
30-year generation	3,63,85,276 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 3,63,85,276 units, avoiding 29,835.9 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹3.23 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 850 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

900 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

900 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

Net investment ₹3.26 Cr after input tax credit, generating 13,79,700 units in year one against an energy charge of ₹7.65 per unit.

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹6.58 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	900 kWp
Year 1 generation	13,79,700 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹98.79 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹3.42 Cr
GST at 5% — recovered as ITC	₹16.29 L
Net capitalised cost	₹3.26 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹81.98 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	3,25,71,428	1,30,28,571	1,95,42,857	32,79,031
2	1,95,42,857	78,17,143	1,17,25,714	19,67,419
3	1,17,25,715	46,90,286	70,35,429	11,80,451
4	70,35,428	28,14,171	42,21,257	7,08,271
5	42,21,257	16,88,503	25,32,754	4,24,962
6	25,32,755	10,13,102	15,19,653	2,54,978
7	15,19,653	6,07,861	9,11,792	1,52,986
8	9,11,792	3,64,717	5,47,075	91,792
9	5,47,075	2,18,830	3,28,245	55,075
10	3,28,245	1,31,298	1,96,947	33,045
11	1,96,947	78,779	1,18,168	19,827
12	1,18,168	47,267	70,901	11,896
Tax shield, years 1-12				81,79,733

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

900 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	13,79,700	98,78,652	—	1,30,28,571	(7,92,772)	1,06,71,424	1,06,71,424
2	13,72,802	1,02,22,429	5,40,000	78,17,143	4,69,455	92,12,974	1,98,84,398
3	13,65,937	1,05,78,170	5,40,000	46,90,286	13,45,955	86,92,214	2,85,76,612
4	13,59,108	1,09,46,290	5,40,000	28,14,171	19,10,784	84,95,506	3,70,72,118
5	13,52,312	1,13,27,221	5,40,000	16,88,503	22,89,965	84,97,255	4,55,69,373
6	13,45,551	1,17,21,408	5,40,000	10,13,102	25,59,159	86,22,249	5,41,91,622
7	13,38,823	1,21,29,313	5,40,000	6,07,861	27,63,812	88,25,501	6,30,17,123
8	13,32,129	1,25,51,413	5,40,000	3,64,717	29,31,241	90,80,173	7,20,97,296
9	13,25,468	1,29,88,202	5,40,000	2,18,830	30,77,888	93,70,314	8,14,67,610
10	13,18,841	1,34,40,192	5,40,000	1,31,298	32,13,675	96,86,517	9,11,54,127
11	13,12,247	1,39,07,910	5,40,000	78,779	33,44,609	1,00,23,302	10,11,77,429
12	13,05,685	1,43,91,906	5,40,000	47,267	34,74,351	1,03,77,554	11,15,54,983
13	12,99,157	1,48,92,744	5,40,000	28,360	36,05,161	1,07,47,583	12,23,02,566
14	12,92,661	1,54,11,012	5,40,000	17,016	37,38,454	1,11,32,558	13,34,35,124
15	12,86,198	1,59,47,315	5,40,000	10,210	38,75,143	1,15,32,171	14,49,67,295
16	12,79,767	1,65,02,281	5,40,000	6,126	40,15,845	1,19,46,436	15,69,13,731
17	12,73,368	1,70,76,561	5,40,000	3,676	41,60,997	1,23,75,564	16,92,89,295
18	12,67,001	1,76,70,825	5,40,000	2,205	43,10,931	1,28,19,894	18,21,09,189
19	12,60,666	1,82,85,770	5,40,000	1,323	44,65,922	1,32,79,847	19,53,89,036
20	12,54,363	1,89,22,115	5,40,000	794	46,26,211	1,37,55,904	20,91,44,940
21	12,48,091	1,95,80,604	5,40,000	476	47,92,019	1,42,48,585	22,33,93,525
22	12,41,851	2,02,62,009	5,40,000	286	49,63,563	1,47,58,446	23,81,51,971
23	12,35,641	2,09,67,127	5,40,000	171	51,41,056	1,52,86,071	25,34,38,042
24	12,29,463	2,16,96,783	5,40,000	103	53,24,713	1,58,32,070	26,92,70,112
25	12,23,316	2,24,51,831	5,40,000	62	55,14,754	1,63,97,077	28,56,67,189
26	12,17,199	2,32,33,155	5,40,000	37	57,11,404	1,69,81,751	30,26,48,940
27	12,11,113	2,40,41,669	5,40,000	22	59,14,894	1,75,86,774	32,02,35,714
28	12,05,058	2,48,78,319	5,40,000	13	61,25,465	1,82,12,854	33,84,48,568
29	11,99,032	2,57,44,084	5,40,000	8	63,43,362	1,88,60,722	35,73,09,290
30	11,93,037	2,66,39,978	5,40,000	5	65,68,841	1,95,31,137	37,68,40,427
Total	3,85,25,585	50,82,87,288	1,56,60,000	3,25,71,421	11,57,86,857	37,68,40,427	37,68,40,427

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹6.58 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹37.68 Cr
30-year generation	3,85,25,586 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 3,85,25,586 units, avoiding 31,591.0 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹3.42 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 900 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

950 kWp

PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

950 KWP GRID-TIED SOLAR PLANT

Net investment ₹3.44 Cr after input tax credit, generating 14,56,350 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹6.94 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	950 kWp
Year 1 generation	14,56,350 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹104.27 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹3.61 Cr
GST at 5% — recovered as ITC	₹17.19 L
Net capitalised cost	₹3.44 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹86.53 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	3,43,80,952	1,37,52,381	2,06,28,571	34,61,199
2	2,06,28,572	82,51,429	1,23,77,143	20,76,720
3	1,23,77,143	49,50,857	74,26,286	12,46,032
4	74,26,285	29,70,514	44,55,771	7,47,619
5	44,55,772	17,82,309	26,73,463	4,48,572
6	26,73,463	10,69,385	16,04,078	2,69,143
7	16,04,078	6,41,631	9,62,447	1,61,486
8	9,62,447	3,84,979	5,77,468	96,892
9	5,77,468	2,30,987	3,46,481	58,135
10	3,46,480	1,38,592	2,07,888	34,881
11	2,07,888	83,155	1,24,733	20,928
12	1,24,733	49,893	74,840	12,557
Tax shield, years 1-12				86,34,162

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

950 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	14,56,350	1,04,27,466	—	1,37,52,381	(8,36,815)	1,12,64,281	1,12,64,281
2	14,49,068	1,07,90,342	5,70,000	82,51,429	4,95,536	97,24,806	2,09,89,087
3	14,41,823	1,11,65,846	5,70,000	49,50,857	14,20,731	91,75,115	3,01,64,202
4	14,34,614	1,15,54,417	5,70,000	29,70,514	20,16,939	89,67,478	3,91,31,680
5	14,27,441	1,19,56,511	5,70,000	17,82,309	24,17,186	89,69,325	4,81,01,005
6	14,20,304	1,23,72,597	5,70,000	10,69,385	27,01,335	91,01,263	5,72,02,268
7	14,13,202	1,28,03,164	5,70,000	6,41,631	29,17,357	93,15,807	6,65,18,075
8	14,06,136	1,32,48,714	5,70,000	3,84,979	30,94,087	95,84,627	7,61,02,702
9	13,99,105	1,37,09,769	5,70,000	2,30,987	32,48,882	98,90,887	8,59,93,589
10	13,92,110	1,41,86,869	5,70,000	1,38,592	33,92,213	1,02,24,656	9,62,18,245
11	13,85,149	1,46,80,572	5,70,000	83,155	35,30,420	1,05,80,152	10,67,98,397
12	13,78,223	1,51,91,456	5,70,000	49,893	36,67,371	1,09,54,085	11,77,52,482
13	13,71,332	1,57,20,119	5,70,000	29,936	38,05,448	1,13,44,671	12,90,97,153
14	13,64,476	1,62,67,179	5,70,000	17,962	39,46,145	1,17,51,033	14,08,48,186
15	13,57,653	1,68,33,277	5,70,000	10,777	40,90,429	1,21,72,848	15,30,21,034
16	13,50,865	1,74,19,075	5,70,000	6,466	42,38,948	1,26,10,127	16,56,31,161
17	13,44,111	1,80,25,259	5,70,000	3,880	43,92,163	1,30,63,096	17,86,94,257
18	13,37,390	1,86,52,538	5,70,000	2,328	45,50,427	1,35,32,110	19,22,26,367
19	13,30,703	1,93,01,646	5,70,000	1,397	47,14,029	1,40,17,617	20,62,43,984
20	13,24,050	1,99,73,343	5,70,000	838	48,83,222	1,45,20,121	22,07,64,105
21	13,17,429	2,06,68,415	5,70,000	503	50,58,243	1,50,40,173	23,58,04,278
22	13,10,842	2,13,87,676	5,70,000	302	52,39,317	1,55,78,359	25,13,82,637
23	13,04,288	2,21,31,967	5,70,000	181	54,26,670	1,61,35,297	26,75,17,934
24	12,97,767	2,29,02,160	5,70,000	109	56,20,531	1,67,11,629	28,42,29,563
25	12,91,278	2,36,99,155	5,70,000	65	58,21,129	1,73,08,026	30,15,37,589
26	12,84,821	2,45,23,886	5,70,000	39	60,28,704	1,79,25,182	31,94,62,771
27	12,78,397	2,53,77,317	5,70,000	23	62,43,500	1,85,63,817	33,80,26,588
28	12,72,005	2,62,60,448	5,70,000	14	64,65,768	1,92,24,679	35,72,51,267
29	12,65,645	2,71,74,311	5,70,000	8	66,95,771	1,99,08,540	37,71,59,807
30	12,59,317	2,81,19,977	5,70,000	5	69,33,777	2,06,16,200	39,77,76,007
Total	4,06,65,894	53,65,25,471	1,65,30,000	3,43,80,945	12,22,19,463	39,77,76,007	39,77,76,007

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹6.94 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹39.78 Cr
30-year generation	4,06,65,896 kWh

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Grid energy charge today		₹7.65

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